

# Mathematical and Physical Foundations of the KT-D1V5 Finite-Volume Discrete Boltzmann Solver

MUSCL reconstruction with a generalized-minmod limiter, SSP-RK3 time integration, and the Sod shock-tube benchmark

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## Abstract

This report explains, without reference to code, the mathematics and physics that underlie the one-dimensional finite-volume discrete Boltzmann solver `d1v5_sodshock_muscl_MC_RK3.m`. The solver advances five discrete velocity populations of the Kataoka–Tsutahara (KT) D1V5 model under a Bhatnagar–Gross–Krook (BGK) relaxation, transports them with an upwind finite-volume flux built on a MUSCL reconstruction limited by the generalized-minmod slope ( $\theta = 1.2$ ), and integrates in time with the three-stage strong-stability-preserving Runge–Kutta method. Each ingredient is derived from first principles: the moment system that fixes the equilibrium, the inverse map from populations to density, velocity, temperature and pressure, the Chapman–Enskog expansion of the discrete model, the exact Riemann solution of the Sod problem, total-variation-diminishing (TVD) limiter theory, the stability of the time integrator, and the error norms used for validation. The analysis yields several results that are specific to this model and were checked numerically: the D1V5 model reproduces the Euler equations exactly at leading order, but its  $\mathcal{O}(\tau)$  dissipation is *not* Navier–Stokes; the contact discontinuity diffuses with a velocity-dependent coefficient  $D_c = \tau(v_1^2 - U^2)(v_2^2 - U^2)/[(b+2)T]$  that vanishes as the flow speed approaches the slow lattice speed; and the BGK stiffness is exactly  $1/\tau$ . A five-grid convergence study ( $N_x = 3125$ – $50,000$ ,  $\tau = 5 \times 10^{-6}$ , CFL = 0.025) together with a matched limiter comparison and a  $\theta$ -sweep is reported and interpreted with these results.

## Contents

|          |                                                               |          |
|----------|---------------------------------------------------------------|----------|
| <b>1</b> | <b>Scope, configuration, and notation</b>                     | <b>3</b> |
| <b>2</b> | <b>Kinetic theory background</b>                              | <b>3</b> |
| 2.1      | The BGK–Boltzmann equation . . . . .                          | 3        |
| 2.2      | Chapman–Enskog expansion and the hydrodynamic limit . . . . . | 4        |
| <b>3</b> | <b>The KT-D1V5 discrete velocity model</b>                    | <b>4</b> |
| 3.1      | Velocity set and internal energy . . . . .                    | 4        |
| 3.2      | Discrete moments and the equilibrium . . . . .                | 5        |
| 3.3      | The first unconstrained moment . . . . .                      | 6        |
| 3.4      | Negative populations . . . . .                                | 6        |
| 3.5      | Recovering the macroscopic state . . . . .                    | 6        |

|           |                                                                               |           |
|-----------|-------------------------------------------------------------------------------|-----------|
| <b>4</b>  | <b>Hydrodynamics of the discrete model</b>                                    | <b>7</b>  |
| 4.1       | Exact conservation . . . . .                                                  | 7         |
| 4.2       | Euler level . . . . .                                                         | 8         |
| 4.3       | First-order Chapman–Enskog analysis . . . . .                                 | 8         |
| 4.4       | What the discrete model gets right and wrong at $\mathcal{O}(\tau)$ . . . . . | 8         |
| 4.5       | Diffusion of a contact discontinuity . . . . .                                | 9         |
| 4.6       | Shock thickness and the kinetic resolution parameter . . . . .                | 10        |
| <b>5</b>  | <b>The Sod shock tube and its exact solution</b>                              | <b>10</b> |
| 5.1       | Problem statement . . . . .                                                   | 10        |
| 5.2       | Exact Riemann solution . . . . .                                              | 11        |
| <b>6</b>  | <b>Finite-volume discretization of the transport term</b>                     | <b>12</b> |
| 6.1       | Cell averages and the semi-discrete equation . . . . .                        | 12        |
| 6.2       | Upwind flux: the exact Riemann solver for each population . . . . .           | 13        |
| 6.3       | Modified equation: numerical diffusion . . . . .                              | 13        |
| 6.4       | MUSCL reconstruction . . . . .                                                | 13        |
| 6.5       | Slope limiters . . . . .                                                      | 14        |
| 6.6       | Boundary treatment . . . . .                                                  | 15        |
| <b>7</b>  | <b>Time integration</b>                                                       | <b>16</b> |
| 7.1       | SSP-RK3 . . . . .                                                             | 16        |
| 7.2       | Linear stability . . . . .                                                    | 16        |
| 7.3       | Time-step selection . . . . .                                                 | 16        |
| 7.4       | Stage checks and fallback . . . . .                                           | 17        |
| <b>8</b>  | <b>Error measures and what to expect from them</b>                            | <b>17</b> |
| 8.1       | Definitions . . . . .                                                         | 17        |
| 8.2       | Expected rates for discontinuous solutions . . . . .                          | 18        |
| <b>9</b>  | <b>Results</b>                                                                | <b>19</b> |
| 9.1       | Run summary and conservation . . . . .                                        | 19        |
| 9.2       | The computed solution . . . . .                                               | 20        |
| 9.3       | Grid convergence . . . . .                                                    | 20        |
| 9.4       | Error by wave: which feature limits convergence . . . . .                     | 23        |
| 9.5       | The velocity error and the sub-cell position of the shock . . . . .           | 24        |
| 9.6       | The role of the finite relaxation time . . . . .                              | 26        |
| 9.7       | Limiter comparison . . . . .                                                  | 27        |
| 9.8       | The $\theta$ sweep . . . . .                                                  | 28        |
| <b>10</b> | <b>Summary</b>                                                                | <b>28</b> |
| 10.1      | How the solver works, in one paragraph . . . . .                              | 28        |
| 10.2      | Principal findings . . . . .                                                  | 29        |
| 10.3      | Limitations and recommended next steps . . . . .                              | 30        |
| <b>A</b>  | <b>Reproducibility</b>                                                        | <b>30</b> |

# 1 Scope, configuration, and notation

The solver computes the classical Sod shock tube [19] using a *finite-volume discrete Boltzmann method* (FVDBM). It does not evolve the Euler variables  $(\rho, \rho u, E)$  directly. Instead it evolves five scalar fields  $f_i(x, t)$ , one per discrete molecular velocity  $c_i$ , whose velocity moments are the Euler variables. The macroscopic gas dynamics emerges from two mechanisms acting on the populations:

1. **free streaming**: each  $f_i$  is advected at its own constant speed  $c_i$  (a linear hyperbolic problem), and
2. **collisions**: each  $f_i$  relaxes toward a local equilibrium  $f_i^{\text{eq}}(\rho, u, T)$  at rate  $1/\tau$  (a nonlinear but local source).

Discrete Boltzmann models of this kind are used for compressible flow because the same populations that reproduce the Euler or Navier–Stokes equations also carry information about thermodynamic non-equilibrium [5, 26]. Finite-volume transport of the populations decouples the velocity lattice from the spatial mesh, which is what makes unstructured and body-fitted grids possible [25]. Table 1 lists the configuration analysed in this report. All quantities are nondimensional with the specific gas constant set to one, so the ideal-gas law reads  $p = \rho T$  and the sound speed is  $a = \sqrt{\gamma T}$ .

**Table 1:** Solver configuration used throughout this report (the live solver and convergence driver of September 18, 2026).

| Quantity                     | Symbol               | Value                                               |
|------------------------------|----------------------|-----------------------------------------------------|
| Specific-heat ratio          | $\gamma$             | 1.4                                                 |
| Degrees-of-freedom parameter | $b = 2/(\gamma - 1)$ | 5                                                   |
| BGK relaxation time          | $\tau$               | $5 \times 10^{-6}$                                  |
| Discrete speeds              | $v_1, v_2$           | 1, 3                                                |
| Internal-energy parameter    | $\eta_0$             | 1.75                                                |
| Domain, diaphragm            | $L_x, x_0$           | 20, 10                                              |
| Final time                   | $t_{\text{end}}$     | 0.15                                                |
| Limiter                      | generalized minmod   | $\theta = 1.20$                                     |
| Time integrator              | SSP-RK3              | $\Delta t = \min(\text{CFL } \Delta x/v_2, \tau/4)$ |
| CFL number                   | CFL                  | 0.025                                               |
| Convergence grids            | $N_x$                | 3125, 6250, 12,500, 25,000, 50,000                  |
| Boundary condition           | —                    | zero-gradient (transmissive)                        |
| Positivity repair            | —                    | off                                                 |

**Claim boundary.** Everything in this report concerns the one-dimensional Sod problem. The analysis explains how the method works and what the measured errors mean; it does not establish grid independence, a formal order of accuracy on a smooth manufactured solution, or behaviour in two dimensions.

## 2 Kinetic theory background

### 2.1 The BGK–Boltzmann equation

Let  $f(x, \xi, \eta, t)$  be a one-dimensional velocity distribution in which  $\xi$  is the molecular velocity along  $x$  and  $\eta$  collects the remaining  $b - 1$  degrees of freedom (the two transverse translational ones and the rotational ones of a diatomic gas), treated as a single internal variable. The Boltzmann

equation with the BGK collision model [1] is

$$\frac{\partial f}{\partial t} + \xi \frac{\partial f}{\partial x} = -\frac{1}{\tau}(f - f^{\text{eq}}), \quad (1)$$

where  $\tau$  is a relaxation time and  $f^{\text{eq}}$  is the local Maxwellian that has the same density, momentum and energy as  $f$ :

$$f^{\text{eq}} = \frac{\rho}{(2\pi T)^{b/2}} \exp\left[-\frac{(\xi - u)^2 + \eta^2}{2T}\right]. \quad (2)$$

The macroscopic fields are velocity moments of  $f$ ,

$$\rho = \int f \, d\Xi, \quad \rho u = \int \xi f \, d\Xi, \quad \rho(bT + u^2) = \int (\xi^2 + \eta^2) f \, d\Xi, \quad (3)$$

with  $d\Xi = d\xi \, d\eta$ . The quantities  $\psi \in \{1, \xi, \frac{1}{2}(\xi^2 + \eta^2)\}$  are *collision invariants*: because  $f^{\text{eq}}$  shares these moments with  $f$ ,  $\int \psi(f^{\text{eq}} - f) \, d\Xi = 0$ , and multiplying (1) by  $\psi$  and integrating yields exact conservation laws for mass, momentum and total energy  $E = \frac{1}{2}\rho(bT + u^2)$ . The internal energy per unit mass is  $e = bT/2$ , so  $c_v = b/2$ ,  $c_p = (b+2)/2$  and  $\gamma = (b+2)/b$ ;  $b = 5$  gives  $\gamma = 1.4$ .

## 2.2 Chapman–Enskog expansion and the hydrodynamic limit

When  $\tau$  is small compared with the macroscopic time scale (small Knudsen number  $\text{Kn} = \tau a/L$ ), the Chapman–Enskog procedure [3, 4] expands

$$f = f^{\text{eq}} + \tau f^{(1)} + \tau^2 f^{(2)} + \dots, \quad \partial_t = \partial_{t_0} + \tau \partial_{t_1} + \dots. \quad (4)$$

At  $\mathcal{O}(1)$  the moment equations are the Euler equations,

$$\partial_t \rho + \partial_x(\rho u) = 0, \quad (5)$$

$$\partial_t(\rho u) + \partial_x(\rho u^2 + p) = 0, \quad (6)$$

$$\partial_t E + \partial_x[u(E + p)] = 0, \quad E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho u^2. \quad (7)$$

At  $\mathcal{O}(\tau)$ ,  $f^{(1)} = -(\partial_{t_0} + \xi \partial_x)f^{\text{eq}}$ , which adds a viscous stress and a heat flux. For the continuous BGK gas with  $b$  degrees of freedom in one-dimensional flow these are

$$\sigma^{\text{BGK}} = 2\mu \left(1 - \frac{1}{b}\right) \partial_x u, \quad q^{\text{BGK}} = -\kappa \partial_x T, \quad \mu = p\tau, \quad \kappa = \frac{b+2}{2} p\tau, \quad (8)$$

so the Prandtl number is one. The Euler equations are recovered as  $\tau \rightarrow 0$ . Section 4 repeats this analysis for the discrete model and shows that the discrete model reproduces the  $\mathcal{O}(1)$  level exactly but not the  $\mathcal{O}(\tau)$  level.

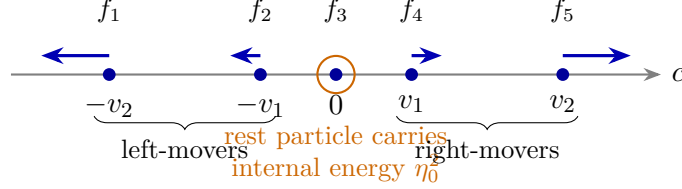
## 3 The KT-D1V5 discrete velocity model

### 3.1 Velocity set and internal energy

The discrete velocity model replaces the continuum  $(\xi, \eta)$  by five particles (fig. 1):

$$c_i \in \{-v_2, -v_1, 0, v_1, v_2\}, \quad \eta_i^2 \in \{0, 0, \eta_0^2, 0, 0\}, \quad i = 1, \dots, 5, \quad (9)$$

with  $v_1 = 1$ ,  $v_2 = 3$  and  $\eta_0 = 1.75$ . Following Kataoka and Tsutahara [9, 10], the entire internal (non- $x$ ) energy of the gas is attached to the rest particle through the parameter  $\eta_0$ . This is what allows a five-velocity model to represent a gas with arbitrary  $\gamma$ : the translational  $x$ -energy is carried by  $c_i^2$  and the remaining  $b - 1$  degrees of freedom by  $\eta_i^2$ . “D1” denotes one spatial dimension; “V5” the five velocities. Unlike a lattice Boltzmann stream-and-collide scheme, the speeds need not map grid nodes to grid nodes, because transport is handled by a finite-volume flux (section 6).



**Figure 1:** The KT-D1V5 velocity set. Arrow lengths are proportional to  $|c_i|$ . Only the rest particle has a nonzero internal-energy coordinate  $\eta_3 = \eta_0$ .

### 3.2 Discrete moments and the equilibrium

The discrete analogues of (3) are

$$\rho = \sum_i f_i, \quad \rho u = \sum_i c_i f_i, \quad 2E = \rho(bT + u^2) = \sum_i (c_i^2 + \eta_i^2) f_i. \quad (10)$$

The equilibrium is required to reproduce, exactly, every moment that appears in the Euler fluxes: the three conserved densities above, the momentum flux, and the energy flux,

$$\sum_i f_i^{\text{eq}} = \rho, \quad \sum_i c_i f_i^{\text{eq}} = \rho u, \quad \sum_i (c_i^2 + \eta_i^2) f_i^{\text{eq}} = \rho(bT + u^2), \quad (11)$$

$$\sum_i c_i^2 f_i^{\text{eq}} = p + \rho u^2, \quad \sum_i c_i (c_i^2 + \eta_i^2) f_i^{\text{eq}} = \rho u[(b+2)T + u^2] = 2u(E + p). \quad (12)$$

Because  $c_3 = 0$ , the last condition involves only the third moment  $M_3 \equiv \sum_i c_i^3 f_i^{\text{eq}}$ . These are five linear conditions on five unknowns, so the equilibrium is *uniquely determined*.

**Proposition 1** (KT-D1V5 equilibrium). *Write  $F_0 = f_3^{\text{eq}}$ ,  $S_k = f_{k+}^{\text{eq}} + f_{k-}^{\text{eq}}$  and  $D_k = f_{k+}^{\text{eq}} - f_{k-}^{\text{eq}}$  for the slow ( $k = 1$ ,  $|c| = v_1$ ) and fast ( $k = 2$ ,  $|c| = v_2$ ) pairs. The unique solution of (11)–(12) is*

$$F_0 = \rho \frac{(b-1)T}{\eta_0^2}, \quad (13)$$

$$S_1 = \rho \frac{v_2^2 - K_2 T - u^2}{v_2^2 - v_1^2}, \quad S_2 = \rho \frac{K_1 T + u^2 - v_1^2}{v_2^2 - v_1^2}, \quad K_k = \frac{(b-1)v_k^2}{\eta_0^2} + 1, \quad (14)$$

$$D_1 = \rho u \frac{v_2^2 - (b+2)T - u^2}{v_1(v_2^2 - v_1^2)}, \quad D_2 = \rho u \frac{(b+2)T + u^2 - v_1^2}{v_2(v_2^2 - v_1^2)}. \quad (15)$$

Equivalently  $f_3^{\text{eq}} = \rho A_0$  and  $f_{k\pm}^{\text{eq}} = \rho(A_k \pm B_k v_k u)$  with

$$\begin{aligned} A_0 &= \frac{(b-1)T}{\eta_0^2}, & A_1 &= \frac{-v_2^2 + K_2 T + u^2}{2(v_1^2 - v_2^2)}, & A_2 &= \frac{-v_1^2 + K_1 T + u^2}{2(v_2^2 - v_1^2)}, \\ B_1 &= \frac{-v_2^2 + (b+2)T + u^2}{2v_1^2(v_1^2 - v_2^2)}, & B_2 &= \frac{-v_1^2 + (b+2)T + u^2}{2v_2^2(v_2^2 - v_1^2)}, \end{aligned} \quad (16)$$

which is exactly the form evaluated by the solver.

*Proof.* The even conditions decouple from the odd ones. The difference of the energy and momentum-flux conditions isolates the rest particle,  $\eta_0^2 F_0 = \rho(bT + u^2) - \rho(T + u^2)$ , that is,  $\eta_0^2 F_0 = \rho(b-1)T$ , giving (13). The remaining even conditions  $S_1 + S_2 = \rho - F_0$  and  $v_1^2 S_1 + v_2^2 S_2 = \rho(T + u^2)$  form a  $2 \times 2$  system whose solution is (14). The odd conditions  $v_1 D_1 + v_2 D_2 = \rho u$  and  $v_1^3 D_1 + v_2^3 D_2 = \rho u[(b+2)T + u^2]$  give (15). Setting  $S_k = 2\rho A_k$  and  $D_k = 2\rho B_k v_k u$  reproduces (16).  $\square$

The equilibrium moments were verified numerically over 2000 random states  $(\rho, u, T)$ : all five residuals are below  $10^{-14}$ .

### 3.3 The first unconstrained moment

The fourth moment is not constrained, and it is what the model “chooses” for the flux of the energy flux. From (14),

$$M_4 \equiv \sum_i c_i^4 f_i^{\text{eq}} = \rho(\alpha_0 + \alpha_1 T + \alpha_2 u^2), \quad \alpha_0 = -v_1^2 v_2^2, \quad \alpha_1 = v_1^2 + v_2^2 + \frac{(b-1)v_1^2 v_2^2}{\eta_0^2}, \quad \alpha_2 = v_1^2 + v_2^2. \quad (17)$$

For the solver’s parameters  $\alpha_0 = -9$ ,  $\alpha_1 = 21.755$  and  $\alpha_2 = 10$ . The continuous Maxwellian would instead give  $\int \xi^2 (\xi^2 + \eta^2) f^{\text{eq}} = \rho[u^4 + (b+5)Tu^2 + (b+2)T^2]$ , which is quadratic in  $T$  and quartic in  $u$ . This mismatch is harmless at the Euler level but controls the dissipative terms (section 4).

### 3.4 Negative populations

The equilibrium is not a probability distribution. From (14), the slow pair is negative whenever  $T > (v_2^2 - u^2)/K_2$  ( $T > 0.706$  at  $u = 0$ ), and the fast pair becomes negative for  $T < (v_1^2 - u^2)/K_1$  ( $T < 0.434$  at  $u = 0$ ). The Sod temperatures lie between 0.711 and 1.141, so the slow populations are negative over almost the entire flow (fig. 2). This is admissible in a discrete Boltzmann method, where the populations are moment carriers rather than particle densities, but it has two consequences. First, no positivity argument can be made at the population level. The solver therefore tests only the recovered  $\rho > 0$  and  $T > 0$ . Second, a population-by-population slope limiter (section 6.5) acts on signed quantities whose extrema have no direct physical meaning.

### 3.5 Recovering the macroscopic state

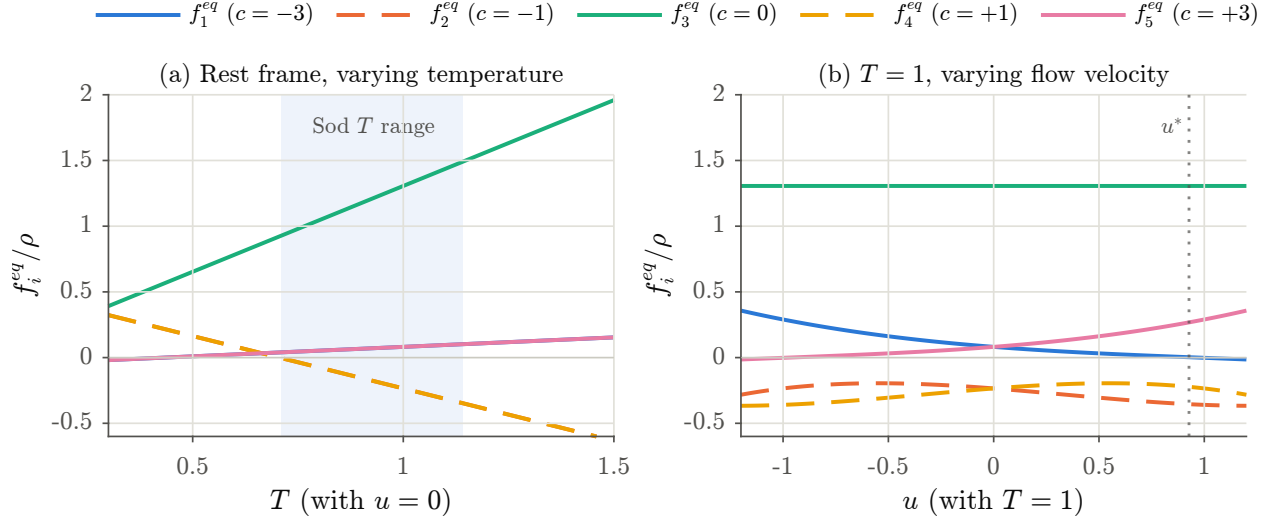
The inverse map from populations to primitive variables follows from (10):

$$\rho = \sum_i f_i, \quad u = \frac{1}{\rho} \sum_i c_i f_i, \quad T = \frac{1}{b} \left[ \frac{1}{\rho} \sum_i (c_i^2 + \eta_i^2) f_i - u^2 \right], \quad p = \rho T. \quad (18)$$

Temperature is the most delicate of these because it is the difference of two  $\mathcal{O}(1)$  quantities. Linearizing (18) gives the sensitivity of  $T$  to a perturbation of one population:

$$\frac{\partial T}{\partial f_i} = \frac{(c_i - u)^2 + \eta_i^2 - bT}{b\rho}. \quad (19)$$

The weights  $\partial T / \partial f_i$  for the Sod states are listed in table 2. In the low-density post-shock state the fast population travelling against the flow has a weight of about 7.3. A reconstruction error in that single population is amplified about sevenfold in  $T$ , while density (weights  $\equiv 1$ ) and pressure respond far less. This is the mechanical reason why temperature is the first variable to show limiter-induced overshoot (section 9.7).



**Figure 2:** KT-D1V5 equilibrium populations per unit density. (a) At  $u = 0$  the pairs  $f_1 = f_5$  and  $f_2 = f_4$  coincide. The shaded band is the range of  $T$  in the Sod solution, where the slow populations are negative. (b) At  $T = 1$ , a nonzero flow velocity splits each pair. Near  $u^* = 0.927$  the fast population moving against the flow ( $f_1$ ) approaches zero.

**Table 2:** Equilibrium populations and temperature sensitivities (19) in the four constant states of the Sod solution.

| State ( $\rho, u, T$ )           |                             | $i = 1 (-3)$ | $i = 2 (-1)$ | $i = 3 (0)$ | $i = 4 (+1)$ | $i = 5 (+3)$ |
|----------------------------------|-----------------------------|--------------|--------------|-------------|--------------|--------------|
| Left (1, 0, 1)                   | $f_i^{eq}$                  | 0.0816       | -0.2347      | 1.306       | -0.2347      | 0.0816       |
|                                  | $\partial T / \partial f_i$ | 0.80         | -0.80        | -0.39       | -0.80        | 0.80         |
| Right (0.125, 0, 0.8)            | $f_i^{eq}$                  | 0.0066       | -0.0094      | 0.1306      | -0.0094      | 0.0066       |
|                                  | $\partial T / \partial f_i$ | 8.0          | -4.8         | -1.5        | -4.8         | 8.0          |
| Star-left (0.426, 0.927, 0.711)  | $f_i^{eq}$                  | 0.0001       | -0.1029      | 0.3959      | 0.0534       | 0.0798       |
|                                  | $\partial T / \partial f_i$ | 5.57         | 0.08         | 0.17        | -1.67        | 0.35         |
| Star-right (0.266, 0.927, 1.141) | $f_i^{eq}$                  | 0.0011       | -0.1089      | 0.3959      | -0.1042      | 0.0817       |
|                                  | $\partial T / \partial f_i$ | 7.32         | -1.50        | -1.34       | -4.29        | -1.06        |

## 4 Hydrodynamics of the discrete model

The discrete BGK equation solved by the program is, for each population,

$$\frac{\partial f_i}{\partial t} + c_i \frac{\partial f_i}{\partial x} = \frac{f_i^{eq}(\rho, u, T) - f_i}{\tau}, \quad i = 1, \dots, 5. \quad (20)$$

Each equation is a linear advection equation with a relaxation source. The nonlinearity is confined to  $f_i^{eq}$ , which depends on all five populations through (18).

### 4.1 Exact conservation

Multiplying (20) by  $\psi_i \in \{1, c_i, \frac{1}{2}(c_i^2 + \eta_i^2)\}$  and summing, the collision term vanishes identically by (11):  $\sum_i \psi_i (f_i^{eq} - f_i) = 0$ . Mass, momentum and energy are therefore conserved by the collision operator exactly, not approximately, and this property survives discretization because the equilibrium is rebuilt from the discrete moments of the current populations.

## 4.2 Euler level

Substituting  $f_i = f_i^{\text{eq}}$  into the summed equations and using (12) gives (5)–(7) exactly. The five conditions (11)–(12) are precisely the conditions needed for this, so the D1V5 model is the minimal discrete velocity model for the one-dimensional Euler equations with adjustable  $\gamma$ .

## 4.3 First-order Chapman–Enskog analysis

With  $f_i = f_i^{\text{eq}} + \tau f_i^{(1)} + \mathcal{O}(\tau^2)$  and  $f_i^{(1)} = -(\partial_{t_0} + c_i \partial_x) f_i^{\text{eq}}$ , the  $\mathcal{O}(\tau)$  corrections to the three fluxes are moments of  $f^{(1)}$ . The time derivatives  $\partial_{t_0}$  are replaced by spatial derivatives using the Euler equations in primitive form,  $\partial_{t_0} \rho = -u \rho_x - \rho u_x$ ,  $\partial_{t_0} u = -u u_x - p_x / \rho$ ,  $\partial_{t_0} p = -u p_x - \gamma p u_x$ .

**Mass.**  $\sum_i c_i f_i^{(1)} = -[\partial_{t_0}(\rho u) + \partial_x(p + \rho u^2)] = 0$ . The mass flux has no dissipative correction.

**Momentum.** Using  $M_2 = p + \rho u^2$  and  $M_3 = \rho u^3 + (b+2)pu$ ,

$$\Pi \equiv \tau \sum_i c_i^2 f_i^{(1)} = -\tau [\partial_{t_0} M_2 + \partial_x M_3] = -\tau(b-1)(\gamma p u_x + u p_x). \quad (21)$$

The momentum equation becomes  $\partial_t(\rho u) + \partial_x(\rho u^2 + p) = \partial_x \sigma$  with  $\sigma = -\Pi = \tau(b-1)(\gamma p u_x + u p_x)$ .

**Energy.** Using (17), the energy-flux correction  $\tilde{q} \equiv \frac{\tau}{2} \sum_i c_i (c_i^2 + \eta_i^2) f_i^{(1)} = -\frac{\tau}{2} [\partial_{t_0} M_3 + \partial_x M_4]$  is

$$\tilde{q} = \frac{\tau}{2} \left[ P(u) \rho_x - (\alpha_1 - (b+2)T - (b+5)u^2) p_x + u(4\rho u^2 + (b+2)(\gamma+1)p - 2\alpha_2 \rho) u_x \right], \quad (22)$$

where

$$P(u) \equiv (v_1^2 - u^2)(v_2^2 - u^2) = -(\alpha_0 + \alpha_2 u^2 - u^4). \quad (23)$$

Both closed forms (21) and (22) were checked against direct numerical differentiation of the actual equilibrium at 500 random states and gradients. The relative discrepancy was below  $3 \times 10^{-9}$  (the finite-difference error).

## 4.4 What the discrete model gets right and wrong at $\mathcal{O}(\tau)$

Table 3 compares (21)–(22) with the continuous BGK result (8). Three differences matter.

1. **Viscosity magnitude.** The coefficient of  $u_x$  in the stress is  $(b-1)\gamma p \tau = 5.6 p \tau$  instead of  $2(1-1/b)p \tau = 1.6 p \tau$ .
2. **Galilean invariance.** The terms  $u p_x$  in  $\sigma$  and the explicit  $u$ -dependence of  $\tilde{q}$  mean that the dissipation depends on the velocity of the frame. A lattice with a finite number of speeds can match only finitely many Maxwellian moments. Galilean invariance of the  $\mathcal{O}(\tau)$  terms needs moments beyond those in (11)–(12) [17].
3. **Spurious density-gradient energy flux.** At  $u = 0$ ,  $\tilde{q} = -\frac{\tau}{2} \rho (\alpha_1 - (b+2)T) T_x + \frac{\tau}{2} [v_1^2 v_2^2 - \alpha_1 T + (b+2)T^2] \rho_x$ . The first term is a heat conduction with  $\kappa_{\text{eff}} = \frac{\tau}{2} \rho (\alpha_1 - (b+2)T)$  ( $\approx 7.4 \rho \tau$  at  $T = 1$  versus the BGK value  $3.5 \rho \tau$ ). The second term is an energy flux driven by density gradients, which has no counterpart in the continuous gas.

The conclusion is that, in this model,  $\tau$  is not a physical viscosity parameter. It is a regularization that vanishes as  $\tau \rightarrow 0$  and leaves the Euler equations. The KT D1V5 model is an *Euler* model [9]; a Navier–Stokes model requires more velocities [10]. This is consistent with using the exact inviscid Sod solution as the reference.

**Table 3:** First-order (in  $\tau$ ) dissipative fluxes: continuous BGK gas versus the KT-D1V5 model, one-dimensional flow.

|                           | Continuous BGK              | KT-D1V5 (derived here)                                                                                             |
|---------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------|
| Mass flux correction      | 0                           | 0                                                                                                                  |
| Viscous stress $\sigma$   | $(3 - \gamma) p \tau u_x$   | $(b - 1) \tau (\gamma p u_x + u p_x)$                                                                              |
| Heat flux at $u = 0$      | $-\frac{b+2}{2} p \tau T_x$ | $-\frac{\tau}{2} \rho (\alpha_1 - (b + 2) T) T_x + \frac{\tau}{2} [v_1^2 v_2^2 - \alpha_1 T + (b + 2) T^2] \rho_x$ |
| Contact diffusivity $D_c$ | $\tau T$                    | $\tau P(U) / [(b + 2) T]$                                                                                          |
| Galilean invariant        | yes                         | no                                                                                                                 |

#### 4.5 Diffusion of a contact discontinuity

A contact discontinuity moves with the flow velocity  $U$  and carries a jump in  $\rho$  and  $T$  at constant  $p$  and  $u$ . At a contact  $u_x = p_x = 0$  to leading order, so  $\Pi = 0$  and (22) reduces to  $\tilde{q} = \frac{\tau}{2} P(U) \rho_x$ . Subtracting the kinetic-energy equation from the energy equation gives the pressure equation  $\frac{b}{2} (\partial_t p + u p_x) + \frac{b+2}{2} p u_x = -\partial_x \tilde{q}$ . Because pressure equilibrates acoustically much faster than the contact diffuses,  $p$  remains uniform. The heat flux therefore produces a small dilatation  $u_x = -2\partial_x \tilde{q} / [(b + 2)p]$ , and the continuity equation  $D\rho/Dt = -\rho u_x$  becomes a diffusion equation,

$$\frac{D\rho}{Dt} = D_c \rho_{xx}, \quad D_c = \frac{\tau (v_1^2 - U^2)(v_2^2 - U^2)}{(b + 2) T}. \quad (24)$$

For the continuous BGK gas the same argument gives  $D_c = \tau T$ . Two features of (24) are notable (fig. 3):

- The diffusivity depends on the contact velocity. At the Sod contact,  $U = u^* = 0.9275$ , the factor  $P(U)/P(0) = 0.126$ . The contact therefore diffuses about eight times more slowly than it would at rest.
- For  $v_1 < |U| < v_2$ ,  $D_c < 0$ . The discrete model is then *anti-diffusive* at contacts, and the  $\mathcal{O}(\tau)$  system is ill-posed. The KT-D1V5 model with  $v_1 = 1$  can therefore represent a contact only if  $|U| < v_1$ . The Sod problem, with  $u^*/v_1 = 0.93$ , sits close to this limit.

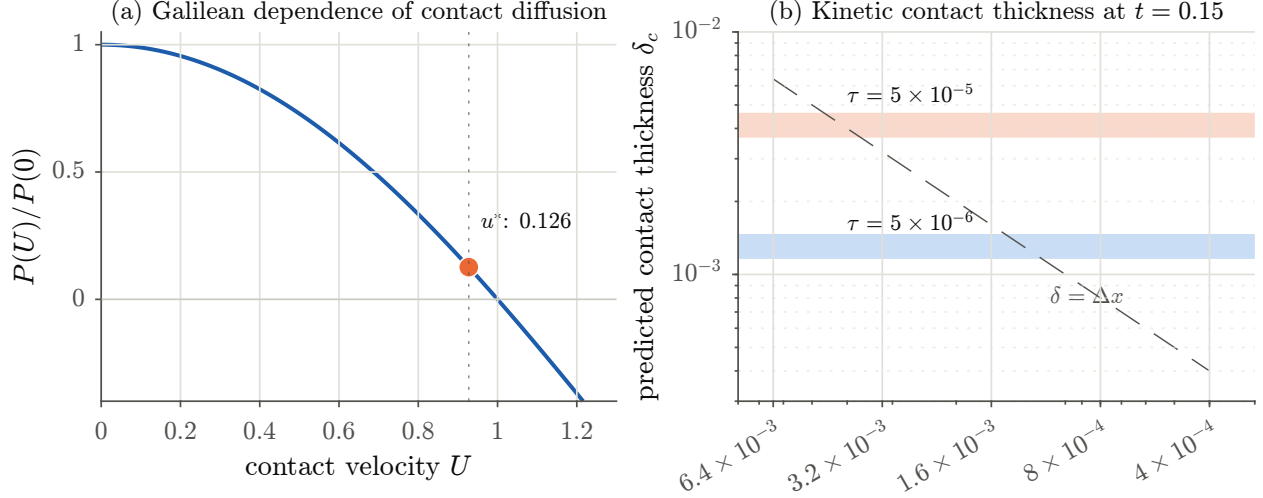
Starting from a step, the solution of (24) is an error function of width  $w = \sqrt{4D_c t}$ . Its maximum-gradient thickness,  $\delta_c \equiv ||[\rho]| / \max |\rho_x|$ , is

$$\delta_c^{\text{kin}} = \sqrt{4\pi D_c t}. \quad (25)$$

Using  $T \in [T_L^*, T_R^*] = [0.711, 1.141]$  and  $t = 0.15$  gives  $\delta_c^{\text{kin}} \in [1.2 \times 10^{-3}, 1.5 \times 10^{-3}]$  for  $\tau = 5 \times 10^{-6}$ . This *kinetic floor* is set by the model, not by the grid. Even an exact solution of (20) would show a contact of this thickness. An error-function contact of width  $w$  and jump  $J$  contributes

$$\int (\rho - \rho_{\text{Euler}})^2 dx = J^2 w \frac{2 - \sqrt{2}}{2\sqrt{\pi}}, \quad (26)$$

to the squared error. With  $J = \rho_L^* - \rho_R^* = 0.1607$ , this sets a floor of  $3.7 \times 10^{-4}$ – $4.2 \times 10^{-4}$  on the global RMS density error defined in section 8. Grid refinement cannot remove it at fixed  $\tau$ .



**Figure 3:** (a) Velocity dependence of the model's contact diffusivity (24). (b) Predicted kinetic contact thickness (25) at  $t = 0.15$  for the present  $\tau$  and for the earlier  $\tau = 5 \times 10^{-5}$ , compared with the cell size of the five convergence grids. The kinetic thickness is less than about four cells even on the finest grid, so it is only marginally resolved.

#### 4.6 Shock thickness and the kinetic resolution parameter

Across the right-moving shock,  $u_x$  and  $p_x$  are both negative, so the two terms of  $\sigma$  add. Balancing the stress against the momentum flux through the shock,  $\dot{m}[u] \sim \mu_{\text{eff}}[u]/\delta_s$  with  $\dot{m} = \rho_R S$ , gives

$$\delta_s \sim \frac{\mu_{\text{eff}}}{\rho_R S}, \quad \mu_{\text{eff}} \approx (b-1)\tau \left( \gamma \bar{p} + \bar{u} \frac{[p]}{[u]} \right) \approx 1.5\tau, \quad (27)$$

so  $\delta_s \approx 7\tau \approx 3.5 \times 10^{-5}$ . This is an order of magnitude smaller than the finest cell ( $\Delta x = 4 \times 10^{-4}$ ). The physical shock structure is therefore never resolved, and the captured shock thickness is set by the numerical scheme. A convenient single measure is the *kinetic resolution parameter*

$$R_\tau = \frac{\Delta x}{v_2 \tau}, \quad (28)$$

the number of fast-particle relaxation lengths per cell. It equals 427, 213, 107, 53 and 27 on the five grids. The solver thus always operates in the under-resolved-relaxation (hydrodynamic) regime. The one exception is the contact, where kinetic diffusion accumulates as  $\sqrt{\tau t}$  rather than scaling with  $\tau$  alone.

## 5 The Sod shock tube and its exact solution

### 5.1 Problem statement

The initial data are two constant states separated by a diaphragm at  $x_0 = 10$ :

$$(\rho, u, p)(x, 0) = \begin{cases} (1, 0, 1), & x < x_0, \\ (0.125, 0, 0.1), & x > x_0. \end{cases} \quad (29)$$

The kinetic solver is initialized at local equilibrium,  $f_i(x, 0) = f_i^{\text{eq}}(\rho, u, T)$ , so no initial non-equilibrium layer is imposed.

## 5.2 Exact Riemann solution

The Euler Riemann problem is self-similar in  $\xi = (x - x_0)/t$  and consists of a left wave, a contact discontinuity moving at  $u^*$ , and a right wave, separating two “star” states with common  $p^*$  and  $u^*$  [21]. The star pressure is the root of

$$\mathcal{F}(p) \equiv f_L(p) + f_R(p) + (u_R - u_L) = 0, \quad (30)$$

where, for  $K \in \{L, R\}$ ,

$$f_K(p) = \begin{cases} (p - p_K) \left[ \frac{A_K}{p + B_K} \right]^{1/2}, & p > p_K \text{ (shock)}, \\ \frac{2a_K}{\gamma - 1} \left[ \left( \frac{p}{p_K} \right)^{\frac{\gamma-1}{2\gamma}} - 1 \right], & p \leq p_K \text{ (rarefaction)}, \end{cases} \quad (31)$$

with  $A_K = 2/[(\gamma+1)\rho_K]$  and  $B_K = p_K(\gamma-1)/(\gamma+1)$ . The shock branch follows from the Rankine–Hugoniot conditions  $[\rho(u-S)] = 0$ ,  $[\rho u(u-S) + p] = 0$ ,  $[E(u-S) + pu] = 0$ . The rarefaction branch follows from constancy of the Riemann invariant  $u + 2a/(\gamma-1)$  and of entropy  $p/\rho^\gamma$  through the fan.  $\mathcal{F}$  is monotone and concave, so Newton’s method,  $p^{(k+1)} = p^{(k)} - \mathcal{F}(p^{(k)})/\mathcal{F}'(p^{(k)})$ , converges from  $p^{(0)} = \frac{1}{2}(p_L + p_R)$ . Then  $u^* = \frac{1}{2}(u_L + u_R) + \frac{1}{2}[f_R(p^*) - f_L(p^*)]$ .

For Sod,  $p^* < p_L$  (left rarefaction) and  $p^* > p_R$  (right shock). Inside the fan,  $\xi \in [u_L - a_L, u^* - a_L^*]$ ,

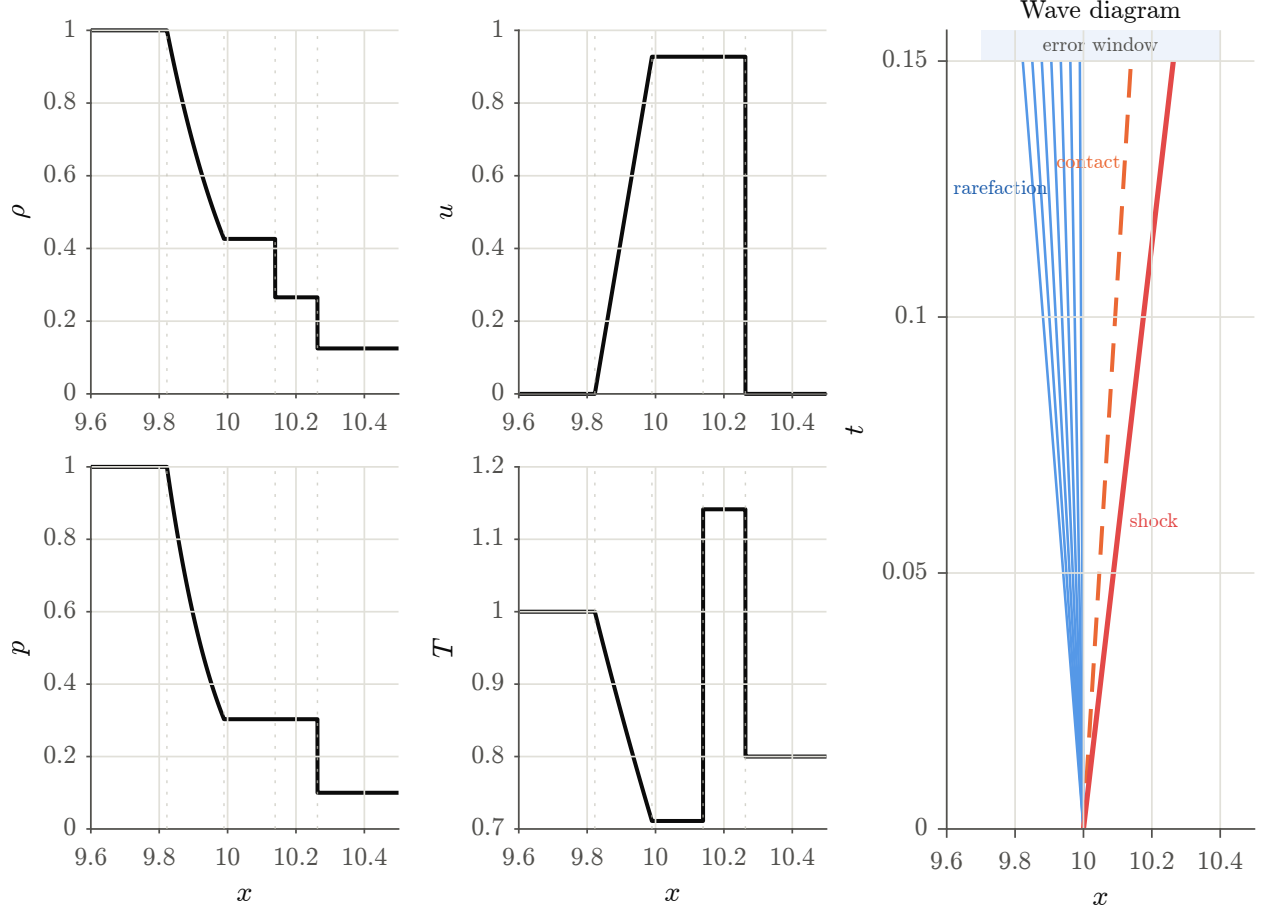
$$u = \frac{2}{\gamma+1}(a_L + \xi), \quad a = \frac{2}{\gamma+1}\left(a_L - \frac{\gamma-1}{2}\xi\right), \quad \rho = \rho_L \left(\frac{a}{a_L}\right)^{\frac{2}{\gamma-1}}, \quad p = p_L \left(\frac{a}{a_L}\right)^{\frac{2\gamma}{\gamma-1}}. \quad (32)$$

Behind the shock,  $\rho_R^* = \rho_R \frac{p^*/p_R + \frac{\gamma-1}{\gamma+1}}{\frac{\gamma-1}{\gamma+1}p^*/p_R + 1}$ , and the shock speed is  $S = a_R \sqrt{\frac{\gamma+1}{2\gamma} \frac{p^*}{p_R} + \frac{\gamma-1}{2\gamma}}$ . The resulting constants (table 4, fig. 4) are used throughout.

**Table 4:** Exact Sod constants ( $\gamma = 1.4$ ) and wave positions at  $t = 0.15$ .

| Quantity          | Value              | Wave             | Speed $\rightarrow$ position $x - x_0$ |
|-------------------|--------------------|------------------|----------------------------------------|
| $p^*$             | 0.303130           | rarefaction head | $-1.18322 \rightarrow -0.17748$        |
| $u^*$             | 0.927453           | rarefaction tail | $-0.07027 \rightarrow -0.01054$        |
| $\rho_L^*, T_L^*$ | 0.426319, 0.711040 | contact          | $0.92745 \rightarrow 0.13912$          |
| $\rho_R^*, T_R^*$ | 0.265574, 1.141416 | shock            | $1.75216 \rightarrow 0.26282$          |
| Shock Mach number | 1.6556             |                  |                                        |

**Why the kinetic solution differs from the Euler reference.** The DBM solution converges to this inviscid solution only in the double limit  $\Delta x \rightarrow 0$  and  $\tau \rightarrow 0$ . At fixed  $\tau$  and  $\Delta x \rightarrow 0$ , it converges instead to the solution of (20). That solution has a smooth contact of thickness (25), a shock of thickness  $\mathcal{O}(\tau)$ , and rounded rarefaction corners. The measured error therefore always contains a model component in addition to the discretization error.



**Figure 4:** Exact Euler solution at  $t = 0.15$  (left, dotted lines mark the four wave positions) and the  $x$ - $t$  wave diagram (right). The shaded strip is the fixed error window  $x_0 - 0.30 \leq x \leq x_0 + 0.40$  used for the wave-region metrics.

## 6 Finite-volume discretization of the transport term

### 6.1 Cell averages and the semi-discrete equation

The domain  $[0, L_x]$  is divided into  $N_x$  cells  $I_j = [x_{j-1/2}, x_{j+1/2}]$  of width  $\Delta x = L_x/N_x$  with centres  $x_j = (j - \frac{1}{2})\Delta x$ . Integrating (20) over  $I_j$  and dividing by  $\Delta x$  gives the exact relation

$$\frac{d\bar{f}_{i,j}}{dt} = -\frac{F_{i,j+1/2} - F_{i,j-1/2}}{\Delta x} + \frac{\bar{f}_i^{\text{eq}} - \bar{f}_{i,j}}{\tau}, \quad F_{i,j+1/2} = c_i f_i(x_{j+1/2}, t), \quad (33)$$

where overbars denote cell averages. The scheme approximates the cell-averaged equilibrium by the equilibrium of the cell averages,  $\bar{f}_i^{\text{eq}}(\bar{\rho}_j, \bar{u}_j, \bar{T}_j)$ . This is a second-order approximation in smooth flow. Writing the right-hand side of (33) as  $\mathcal{L}(\bar{\mathbf{f}})$  gives the method-of-lines system  $d\bar{\mathbf{f}}/dt = \mathcal{L}(\bar{\mathbf{f}})$  that is integrated in time in section 7.

Because the flux enters as a difference of face values, summing over cells telescopes. The total amount of any population, and hence of mass, momentum and energy, changes only through the boundary faces. The finite-volume scheme is conservative by construction.

## 6.2 Upwind flux: the exact Riemann solver for each population

Each population obeys a linear advection equation with constant speed  $c_i$ . The Riemann problem at a face with left and right states  $f^-$  and  $f^+$  has the exact solution “take the upstream value”. The Godunov flux is therefore the upwind flux

$$F_{i,j+1/2} = c_i^+ f_{i,j+1/2}^- + c_i^- f_{i,j+1/2}^+, \quad c^\pm = \frac{1}{2}(c \pm |c|), \quad (34)$$

and  $F = 0$  for the rest particle. Summed over populations, (34) splits the macroscopic flux into contributions from right-movers and left-movers. This is the discrete-velocity analogue of kinetic flux-vector splitting [14]. Upwinding happens separately for each population and needs no characteristic decomposition of the Euler system. The kinetic model provides the wave decomposition automatically.

## 6.3 Modified equation: numerical diffusion

If  $f^\pm$  are the cell averages (first-order upwind), a Taylor expansion of the semi-discrete scheme gives the modified equation [24]

$$\frac{\partial f_i}{\partial t} + c_i \frac{\partial f_i}{\partial x} = \frac{|c_i| \Delta x}{2} \frac{\partial^2 f_i}{\partial x^2} - \frac{c_i \Delta x^2}{6} \frac{\partial^3 f_i}{\partial x^3} + \dots + \frac{f_i^{\text{eq}} - f_i}{\tau}. \quad (35)$$

Taking moments with  $f_i \approx f_i^{\text{eq}}$  shows how the population-level numerical diffusion reaches the macroscopic equations. For example, the continuity equation gains the numerical flux

$$F_\rho^{\text{num}} = -\frac{\Delta x}{2} \partial_x \sum_i |c_i| f_i^{\text{eq}} = -\frac{\Delta x}{2} \partial_x (v_1 S_1 + v_2 S_2) \neq 0. \quad (36)$$

First-order kinetic upwinding therefore diffuses mass, momentum and energy with an effective coefficient of order  $v_2 \Delta x / 2$ . Comparing this with the model dissipation of order  $\tau \cdot \mathcal{O}(1)$  reproduces the resolution parameter (28): numerical dissipation dominates whenever  $R_\tau \gg 1$ . This is why a second-order reconstruction is essential.

## 6.4 MUSCL reconstruction

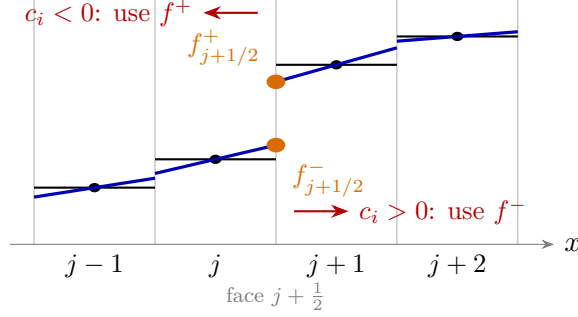
MUSCL [23] replaces the piecewise-constant data by a piecewise-linear reconstruction inside each cell,

$$f_i(x) \approx \bar{f}_{i,j} + \sigma_{i,j} \frac{x - x_j}{\Delta x}, \quad x \in I_j, \quad (37)$$

so the two face states are

$$f_{i,j+1/2}^- = \bar{f}_{i,j} + \frac{1}{2} \sigma_{i,j}, \quad f_{i,j+1/2}^+ = \bar{f}_{i,j+1} - \frac{1}{2} \sigma_{i,j+1}. \quad (38)$$

With the unlimited central slope  $\sigma_j = \frac{1}{2}(\bar{f}_{j+1} - \bar{f}_{j-1})$ , the  $\mathcal{O}(\Delta x)$  diffusion term cancels, and the leading truncation error becomes the dispersive term  $\mathcal{O}(\Delta x^2) \partial_x^3 f$  (Fromm’s scheme). Unlimited reconstruction creates new extrema at discontinuities (Godunov’s theorem [13]), so the slope is limited.



**Figure 5:** MUSCL reconstruction and upwind face selection for one population. Black: cell averages; blue: limited linear reconstructions; orange: the two face states at  $x_{j+1/2}$ . The sign of  $c_i$  selects which state enters the flux.

## 6.5 Slope limiters

Let  $\Delta_{j-1/2} = \bar{f}_j - \bar{f}_{j-1}$  and  $\Delta_{j+1/2} = \bar{f}_{j+1} - \bar{f}_j$ . With

$$\text{minmod}(a_1, \dots, a_m) = \begin{cases} \min_k a_k & \text{if all } a_k > 0, \\ \max_k a_k & \text{if all } a_k < 0, \\ 0 & \text{otherwise,} \end{cases}$$

the three limiters used in the project are

$$\text{minmod: } \sigma_j = \text{minmod}(\Delta_{j-1/2}, \Delta_{j+1/2}), \quad (39)$$

$$\text{MC [22]: } \sigma_j = \text{minmod}(2\Delta_{j-1/2}, \frac{1}{2}(\Delta_{j-1/2} + \Delta_{j+1/2}), 2\Delta_{j+1/2}), \quad (40)$$

$$\text{generalized minmod [11]: } \sigma_j = \text{minmod}(\theta\Delta_{j-1/2}, \frac{1}{2}(\Delta_{j-1/2} + \Delta_{j+1/2}), \theta\Delta_{j+1/2}), \quad 1 \leq \theta \leq 2. \quad (41)$$

The generalized family contains the other two. At  $\theta = 2$  it is MC. At  $\theta = 1$  the centred candidate always lies between the two one-sided candidates, so it is never the minimum in magnitude, and the family reduces to minmod. The solver uses  $\theta = 1.2$ .

**Sweby form.** Dividing by  $\Delta_{j+1/2}$  and writing  $r_j = \Delta_{j-1/2}/\Delta_{j+1/2}$  gives  $\sigma_j = \phi(r_j)\Delta_{j+1/2}$  with

$$\phi_\theta(r) = \max\left[0, \min\left(\theta r, \frac{1+r}{2}, \theta\right)\right]. \quad (42)$$

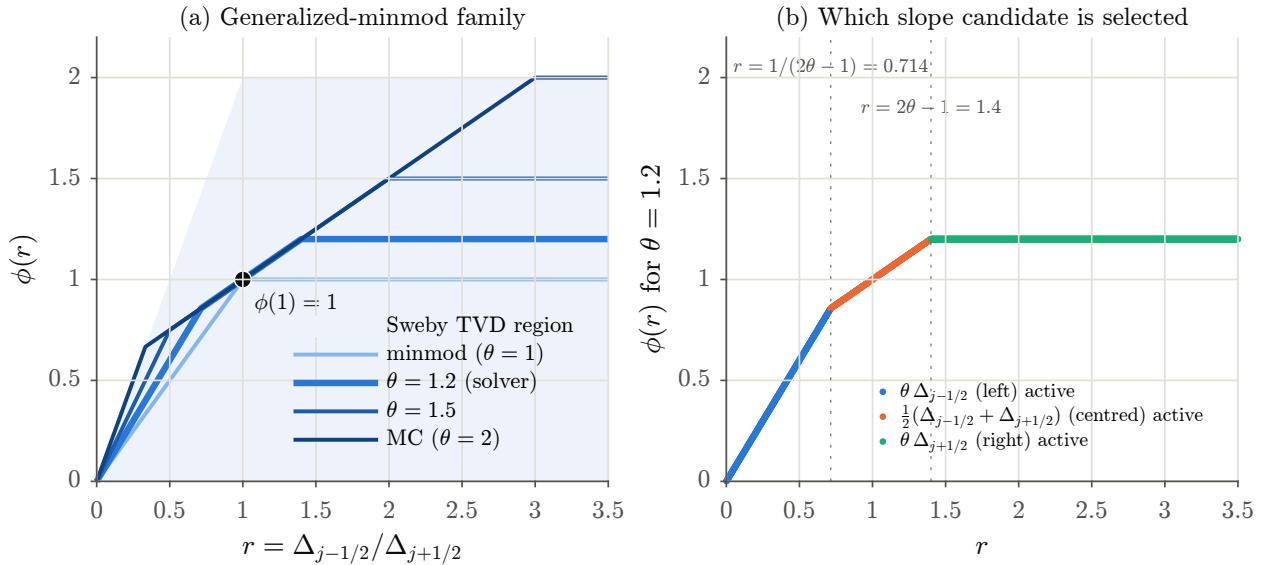
For  $r \leq 0$  (a local extremum),  $\phi = 0$  and the scheme is first order locally. For  $r > 0$  the selected candidate changes at  $r = 1/(2\theta - 1)$  and  $r = 2\theta - 1$ . For  $\theta = 1.2$  the left candidate is active for  $r < 0.714$ , the centred candidate for  $0.714 < r < 1.4$ , and the right candidate beyond that (fig. 6b). Thus  $\theta = 1.2$  differs from minmod only in the window  $0.714 < r < 1.4$  and in the caps  $\theta r$  and  $\theta$ , which allow slopes up to 20% steeper than a one-sided difference. Every member satisfies  $\phi(1) = 1$ , so the reconstruction is exact for linear data and second order in smooth monotone regions. Each member also satisfies the symmetry  $\phi(r)/r = \phi(1/r)$ , so increasing and decreasing data are treated alike.

**Proposition 2** (TVD range of  $\theta$ ). *For a single population with  $c > 0$ , apply the reconstruction (38) with the limiter (42) and advance with forward Euler, using  $\nu = c\Delta t/\Delta x$ . The scheme is total-variation diminishing if  $1 \leq \theta \leq 2$  and  $\nu \leq 2/(2 + \theta)$ .*

*Proof.* The update can be written  $\bar{f}_j^{n+1} = \bar{f}_j - C_{j-1/2} \Delta_{j-1/2}$  with  $C_{j-1/2} = \nu[1 + \frac{1}{2}\phi(r_j)/r_j - \frac{1}{2}\phi(r_{j-1})]$ . By Harten's lemma [8], the scheme is TVD if  $0 \leq C_{j-1/2} \leq 1$ . From (42),  $0 \leq \phi(r) \leq \theta$  and  $0 \leq \phi(r)/r \leq \theta$ . Hence  $\nu(1 - \theta/2) \leq C \leq \nu(1 + \theta/2)$ . The lower bound is non-negative only if  $\theta \leq 2$ , and the upper bound is at most one if  $\nu \leq 2/(2 + \theta)$ . The condition  $\theta \geq 1$  keeps  $\phi$  inside the second-order region of (author?) [20].  $\square$

This shows why  $\theta$  is restricted to  $[1, 2]$ :  $\theta > 2$  breaks the TVD property. For  $\theta = 1.2$  the per-population TVD limit is  $\nu \leq 0.625$ . The solver's largest population Courant number is  $\nu_{\max} = v_2 \Delta t / \Delta x \leq 0.0094$ , nearly seventy times smaller. Two caveats apply:

- TVD holds for each population's *transport*. The collision term couples the populations through  $f^{\text{eq}}(\rho, u, T)$ , so TVD of the full system, or monotonicity of  $\rho$ ,  $u$ ,  $p$  or  $T$ , is not guaranteed.
- Limiting is population by population, while  $T$  is the nonlinear combination (18). A slope that is admissible for every  $f_i$  separately can still create an extremum in  $T$ . This happens most easily through the high-sensitivity populations of table 2.



**Figure 6:** (a) Generalized-minmod limiter functions (42) inside Sweby's TVD region  $0 \leq \phi \leq \min(2r, 2)$ . All members pass through  $\phi(1) = 1$ . (b) For  $\theta = 1.2$ , which of the three candidate slopes is selected, as a function of the smoothness ratio  $r$ .

**History of the limiter choice.** In the project, minmod proved robust but diffusive, and MC proved sharper but produced a temperature overshoot. Two binary hybrids were also tried: one switched per population, and one switched on a shared macroscopic troubled-cell mask. The per-population switch was inconsistent with moment recovery, and the shared mask was threshold-sensitive. The generalized minmod replaces the switch with a single continuous parameter, and its consequences are quantified in section 9.7.

## 6.6 Boundary treatment

The first and last cells copy their interior neighbours after every stage, and the boundary face fluxes copy the adjacent interior face flux. This zero-gradient condition is exact while the domain ends

remain in their undisturbed states. At  $t = 0.15$  the rarefaction head and the shock are more than 9.7 units from the boundaries, so boundary effects are absent (section 9.1).

## 7 Time integration

### 7.1 SSP-RK3

The semi-discrete system  $d\mathbf{f}/dt = \mathcal{L}(\mathbf{f})$ , with  $\mathcal{L} = \text{transport} + \text{collision}$ , is advanced by the three-stage, third-order strong-stability-preserving Runge–Kutta method of Shu and Osher [6, 18]:

$$\begin{aligned} \mathbf{f}^{(1)} &= \mathbf{f}^n + \Delta t \mathcal{L}(\mathbf{f}^n), \\ \mathbf{f}^{(2)} &= \frac{3}{4}\mathbf{f}^n + \frac{1}{4}[\mathbf{f}^{(1)} + \Delta t \mathcal{L}(\mathbf{f}^{(1)})], \\ \mathbf{f}^{n+1} &= \frac{1}{3}\mathbf{f}^n + \frac{2}{3}[\mathbf{f}^{(2)} + \Delta t \mathcal{L}(\mathbf{f}^{(2)})]. \end{aligned} \quad (43)$$

Each stage is a convex combination of forward-Euler steps. Therefore any convex functional that is non-increasing under forward Euler, such as total variation or a maximum norm, remains non-increasing under (43) with the same time-step restriction (SSP coefficient 1). The equivalent Butcher tableau has nodes  $(0, 1, \frac{1}{2})$  and weights  $(\frac{1}{6}, \frac{1}{6}, \frac{2}{3})$ .

### 7.2 Linear stability

For  $dy/dt = \lambda y$ , one step multiplies  $y$  by

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6}, \quad z = \lambda \Delta t, \quad (44)$$

which matches  $e^z$  to third order. The stability region  $|R(z)| \leq 1$  (fig. 7) contains the real interval  $[-2.513, 0]$  and the imaginary interval  $|\text{Im } z| \leq \sqrt{3}$ .

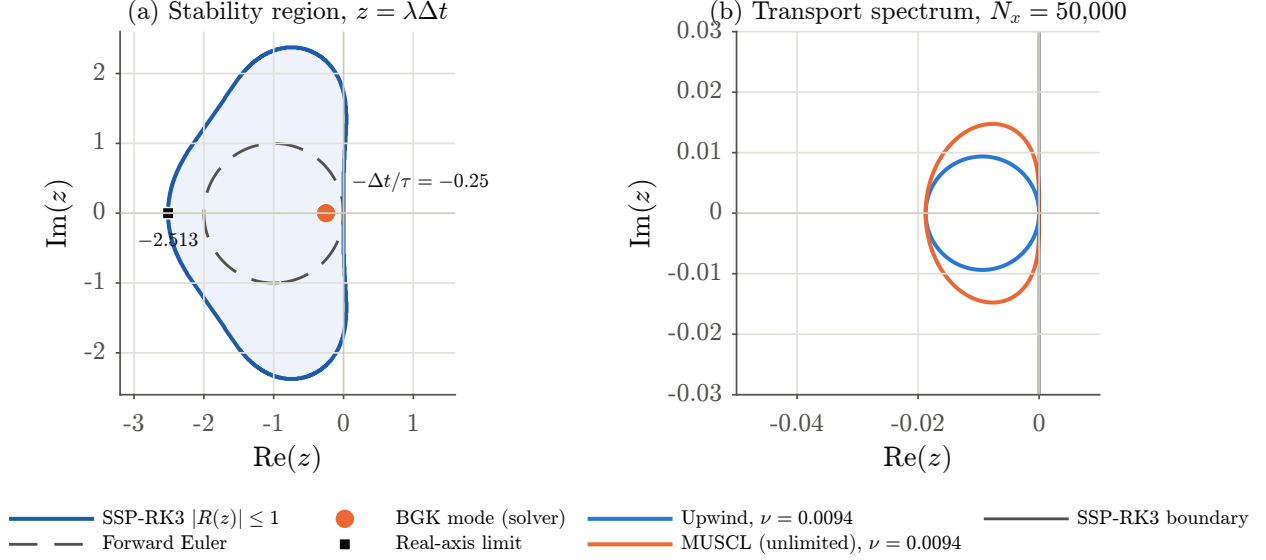
**Collision stiffness.** Linearize the collision term about any state. Its Jacobian is  $\tau^{-1}(\partial f^{\text{eq}}/\partial \mathbf{f} - I)$ . Because  $f^{\text{eq}}$  depends on  $\mathbf{f}$  only through the three conserved moments and reproduces them exactly,  $\partial f^{\text{eq}}/\partial \mathbf{f}$  is a projection with eigenvalues  $\{1, 1, 1, 0, 0\}$  (confirmed numerically). The collision spectrum is therefore exactly  $\{0, 0, 0, -\tau^{-1}, -\tau^{-1}\}$ : the conserved moments do not relax, and the two non-conserved modes relax at rate  $1/\tau$ . Explicit stability requires  $\Delta t \leq 2.513\tau$ . The solver uses  $\Delta t_{\text{col}} = \tau/4$  ( $z = -0.25$ ), for which  $R(-0.25) = 0.778646$  versus  $e^{-0.25} = 0.778801$ . This is a relative decay error of  $2 \times 10^{-4}$  per step, so the relaxation is resolved accurately, not merely stably. The forward-Euler collision step  $f + \frac{\Delta t}{\tau}(f^{\text{eq}} - f)$  is a convex combination of  $f$  and  $f^{\text{eq}}$  for  $\Delta t \leq \tau$ .

**Transport spectrum.** With the unlimited MUSCL slope, the Fourier symbol of the semi-discrete transport operator for a population with  $c > 0$  is  $\lambda(k)\Delta t = -\nu(1 - e^{-ik})(1 + \frac{i}{2}\sin k)$ . For the finest grid ( $\nu = 0.0094$ ) this curve lies deep inside the stability region (fig. 7b).

### 7.3 Time-step selection

The solver takes

$$\Delta t = \min\left(\underbrace{\text{CFL} \frac{\Delta x}{v_2}}_{\Delta t_{\text{adv}}}, \underbrace{\frac{1}{4}\tau}_{\Delta t_{\text{col}}}\right), \quad \nu_{\text{eff}} = \frac{v_2 \Delta t}{\Delta x}. \quad (45)$$



**Figure 7:** (a) SSP-RK3 absolute-stability region with the solver’s BGK eigenvalue  $z = -\Delta t/\tau = -0.25$ . (b) Transport spectra for upwind and unlimited MUSCL reconstruction at the finest grid’s Courant number, near the origin.

With  $\tau = 5 \times 10^{-6}$ ,  $\Delta t_{\text{col}} = 1.25 \times 10^{-6}$ . With  $\text{CFL} = 0.025$ ,  $\Delta t_{\text{adv}}$  ranges from  $5.3 \times 10^{-5}$  ( $N_x = 3125$ ) down to  $3.3 \times 10^{-6}$  ( $N_x = 50,000$ ). The collision limit therefore controls on *every* grid. All five runs use the same  $\Delta t = 1.25 \times 10^{-6}$  and the same number of steps. The effective Courant number  $\nu_{\text{eff}}$  doubles with each refinement, from  $5.9 \times 10^{-4}$  to  $9.4 \times 10^{-3}$ , while the CFL input never becomes active. Two consequences follow:

1. The convergence study is a pure spatial refinement at fixed  $\Delta t$ . It is not a refinement at constant Courant number.
2. Because  $\nu_{\text{eff}} \leq 0.0094$  and  $\Delta t/\tau = 0.25$ , temporal error is expected to be far below spatial error. No separate time-step refinement was run for this report, so this is an expectation rather than a measurement.

## 7.4 Stage checks and fallback

After every stage the solver recovers  $(\rho, T)$  and rejects the step if either is non-positive or non-finite. A rejected step is retried with  $\Delta t/2$ . From the fifth retry onward it uses minmod, and from the ninth first-order upwind. Positivity repair (clipping) is disabled. All runs reported here completed with zero retries and zero fallbacks, so every result is a pure generalized-minmod solution.

## 8 Error measures and what to expect from them

### 8.1 Definitions

For a field  $q$  with numerical cell values  $q_j$  and exact values  $q_j^{\text{ex}}$  evaluated at the cell centres, the solver reports

$$L_1(q) = \frac{1}{N} \sum_j |q_j - q_j^{\text{ex}}|, \quad L_2(q) = \left[ \frac{1}{N} \sum_j (q_j - q_j^{\text{ex}})^2 \right]^{1/2}, \quad (46)$$

---

**Algorithm 1** One accepted time step of the solver

---

```

1: function RHS(f)
2:    $(\rho, u, T) \leftarrow$  moments of f by (18)
3:    $f^{\text{eq}} \leftarrow$  KT-D1V5 equilibrium (16)
4:   for each population  $i$  do
5:     limited slopes  $\sigma_{i,j}$  by (41); face states by (38)
6:     upwind face fluxes by (34); divergence  $(F_{j+1/2} - F_{j-1/2})/\Delta x$ 
7:   end for
8:   return  $-\text{divergence} + (f^{\text{eq}} - \mathbf{f})/\tau$ 
9: end function
10:  $\Delta t \leftarrow$  (45)
11: repeat
12:   three SSP-RK3 stages (43), zero-gradient boundaries after each
13:   if any stage has  $\rho \leq 0$  or  $T \leq 0$ :  $\Delta t \leftarrow \Delta t/2$ , retry (fallback limiters after 4 and 8 retries)
14: until all stages physical

```

---

over all  $N = N_x$  cells (global) or over the  $N_W$  cells in the fixed window  $W = [x_0 - 0.30, x_0 + 0.40]$  (wave region). These are mean and root-mean-square errors. Since  $N\Delta x = L_x$  is fixed, they equal  $L_x^{-1}\|e\|_{L^1}$  and  $L_x^{-1/2}\|e\|_{L^2}$ , so observed orders are unaffected by the normalization. Because the undisturbed part of the domain is 19.56/20 of its length, global values are diluted by a factor of about  $W/L_x = 0.035$  ( $L_1$ ) or  $\sqrt{0.035}$  ( $L_2$ ) relative to wave-region values. The temperature peak excess  $\max_W T - \max_W T^{\text{ex}}$  measures overshoot above the exact post-shock value  $T_R^* = 1.1414$ .

The observed order between two grids and the least-squares order over all grids are

$$p_k = \frac{\ln(E_{k-1}/E_k)}{\ln(\Delta x_{k-1}/\Delta x_k)}, \quad \ln E = \ln C + p \ln \Delta x \quad (\text{fit}), \quad (47)$$

following standard verification practice [15, 16].

## 8.2 Expected rates for discontinuous solutions

The Sod solution is not smooth, so the formal second-order accuracy of MUSCL does not translate into second-order convergence of these norms. Suppose a discontinuity with jump  $J$  is smeared over a thickness  $\delta$ . The error is  $\mathcal{O}(J)$  over a length  $\mathcal{O}(\delta)$ , so

$$L_1 \propto J\delta, \quad L_2 \propto J\delta^{1/2}. \quad (48)$$

The  $L_2$  rate is therefore half the  $L_1$  rate. How  $\delta$  scales with  $\Delta x$  depends on the wave:

- **Shock.** Characteristics converge into a shock, which balances numerical diffusion and keeps  $\delta_s = \mathcal{O}(\Delta x)$  (a fixed number of cells). Hence  $L_1 \propto \Delta x$  and  $L_2 \propto \Delta x^{1/2}$ .
- **Contact.** A contact is linearly degenerate, so nothing counteracts numerical diffusion and it keeps spreading. For a method of order  $m$  the smeared width grows like  $\delta_c \propto (\Delta x^m t)^{1/(m+1)}$  [7, 13]. That gives  $\delta_c \propto \Delta x^{1/2}$  for first order and  $\Delta x^{2/3}$  for second order. The corresponding rates are  $L_1 \propto \Delta x^{2/3}$  and  $L_2 \propto \Delta x^{1/3}$  for MUSCL. The classical results on discontinuous data [2, 12] bound these rates from below.
- **Rarefaction.** The rarefaction is continuous with kinks at its head and tail, so its error converges faster than either discontinuity.

Since the global squared error is the sum of the squared contributions of the three waves, the asymptotic  $L_2$  rate is the *slowest* of the three, which is  $1/3$  from the contact. At fixed  $\tau$  there is also the model floor of section 4.5. Once  $\delta_c^{\text{num}} \lesssim \delta_c^{\text{kin}}$ , the contact error stops decreasing. Heuristically,

$$E_{L_2}^2 \approx C_s \Delta x + C_c [(\delta_c^{\text{num}})^2 + (\delta_c^{\text{kin}})^2]^{1/2} + E_{\text{raref}}^2. \quad (49)$$

Observed global  $L_2$  orders between about  $1/3$  and  $1/2$  are therefore the *correct* behaviour for a second-order shock-capturing method on this problem. They are not a symptom of a defect.

## 9 Results

All runs use the configuration of table 1. The convergence study was executed with `run_d1v5_gminmod_L2_convergence.m`. The limiter comparison and the  $\theta$ -sweep use  $N_x = 6250$  with the same  $\tau$ , CFL and final time.

### 9.1 Run summary and conservation

Table 5 confirms the time-step analysis of section 7.3. Every grid used  $\Delta t = \tau/4$ , the same 120,001 steps (the last is a round-off-sized remainder that lands exactly on  $t_{\text{end}}$ ), and no retries or fallbacks. Table 6 checks the conservation argument of section 4.1. Mass and energy are conserved to round-off relative to the discrete initial condition. The momentum equals  $(p_L - p_R) t_{\text{end}} = 0.135$  to round-off, because the undisturbed boundary states exert the net pressure force  $p_L - p_R$  on the domain.

**Table 5:** Convergence runs: grid spacing, time step, effective Courant number  $\nu_{\text{eff}} = v_2 \Delta t / \Delta x$ , number of steps, and stability events.

| $N_x$  | $\Delta x$           | $\Delta t$            | $\nu_{\text{eff}}$    | steps   | retries | fallbacks |
|--------|----------------------|-----------------------|-----------------------|---------|---------|-----------|
| 3,125  | $6.4 \times 10^{-3}$ | $1.25 \times 10^{-6}$ | $5.86 \times 10^{-4}$ | 120,001 | 0       | 0         |
| 6,250  | $3.2 \times 10^{-3}$ | $1.25 \times 10^{-6}$ | $1.17 \times 10^{-3}$ | 120,001 | 0       | 0         |
| 12,500 | $1.6 \times 10^{-3}$ | $1.25 \times 10^{-6}$ | $2.34 \times 10^{-3}$ | 120,001 | 0       | 0         |
| 25,000 | $8.0 \times 10^{-4}$ | $1.25 \times 10^{-6}$ | $4.69 \times 10^{-3}$ | 120,001 | 0       | 0         |
| 50,000 | $4.0 \times 10^{-4}$ | $1.25 \times 10^{-6}$ | $9.38 \times 10^{-3}$ | 120,001 | 0       | 0         |

**Table 6:** Conservation errors at  $t = 0.15$ :  $\sum_j q_j \Delta x$  minus its expected value (mass and energy: discrete initial value; momentum:  $(p_L - p_R)t$ ).

| $N_x$  | mass                   | momentum               | energy                 |
|--------|------------------------|------------------------|------------------------|
| 3,125  | $-8.2 \times 10^{-13}$ | $-4.1 \times 10^{-13}$ | $-2.3 \times 10^{-12}$ |
| 6,250  | $-7.1 \times 10^{-13}$ | $-3.8 \times 10^{-13}$ | $-1.9 \times 10^{-12}$ |
| 12,500 | $-6.4 \times 10^{-13}$ | $-3.8 \times 10^{-13}$ | $-1.7 \times 10^{-12}$ |
| 25,000 | $-6.0 \times 10^{-13}$ | $-3.7 \times 10^{-13}$ | $-1.5 \times 10^{-12}$ |
| 50,000 | $-5.8 \times 10^{-13}$ | $-3.7 \times 10^{-13}$ | $-1.4 \times 10^{-12}$ |

**Odd grid.**  $N_x = 3125$  is odd, so the cell centre  $x_{1563}$  coincides with the diaphragm. The solver assigns cells with  $x_j < x_0$  to the left state, so that cell receives the right state. The discrete diaphragm therefore sits at  $x_0 - \Delta x/2$ . This is an  $\mathcal{O}(\Delta x)$  initial-condition offset present only on the coarsest grid. It contributes to that grid’s contact position error (table 11) and to the first pairwise orders.

## 9.2 The computed solution

Figure 8 compares the  $N_x = 50,000$  solution with the exact Euler solution, and fig. 9 shows how each wave sharpens under refinement. The profiles are free of oscillations, and the four constant states are reproduced to plotting accuracy. The pointwise error is concentrated in a few cells at each wave. The largest errors occur at the shock in every variable and at the contact in  $\rho$  and  $T$ . The rarefaction head and tail show small corner errors. The largest of these is a slight density undershoot just behind the tail (fig. 9a), which is limited in amplitude and shrinks with  $\Delta x$ . The maximum pointwise error does not decrease with  $\Delta x$ : it stays at  $\approx 0.08$  in  $\rho$  and 0.34–0.63 in  $u$ . This is expected, because an  $\mathcal{O}(1)$  jump spread over a few cells always leaves an  $\mathcal{O}(1)$  error in the cell that straddles it.

## 9.3 Grid convergence

Tables 7, 8 and 10 and fig. 10 give the errors and observed orders. Because the solution outside the error window is exact to round-off, the wave-region  $L_2$  errors (table 9) are the global ones multiplied by  $\approx \sqrt{L_x/0.7} = 5.35$ . They therefore have identical orders.

**Table 7:** Global  $L_1$  errors (mean absolute error over all cells).

| $N_x$  | $\rho$                | $u$                   | $p$                   | $T$                   |
|--------|-----------------------|-----------------------|-----------------------|-----------------------|
| 3,125  | $2.46 \times 10^{-4}$ | $4.14 \times 10^{-4}$ | $2.14 \times 10^{-4}$ | $3.58 \times 10^{-4}$ |
| 6,250  | $1.30 \times 10^{-4}$ | $2.69 \times 10^{-4}$ | $1.09 \times 10^{-4}$ | $2.08 \times 10^{-4}$ |
| 12,500 | $7.23 \times 10^{-5}$ | $1.43 \times 10^{-4}$ | $5.75 \times 10^{-5}$ | $1.17 \times 10^{-4}$ |
| 25,000 | $4.07 \times 10^{-5}$ | $6.29 \times 10^{-5}$ | $3.11 \times 10^{-5}$ | $6.36 \times 10^{-5}$ |
| 50,000 | $2.51 \times 10^{-5}$ | $3.97 \times 10^{-5}$ | $1.88 \times 10^{-5}$ | $3.98 \times 10^{-5}$ |

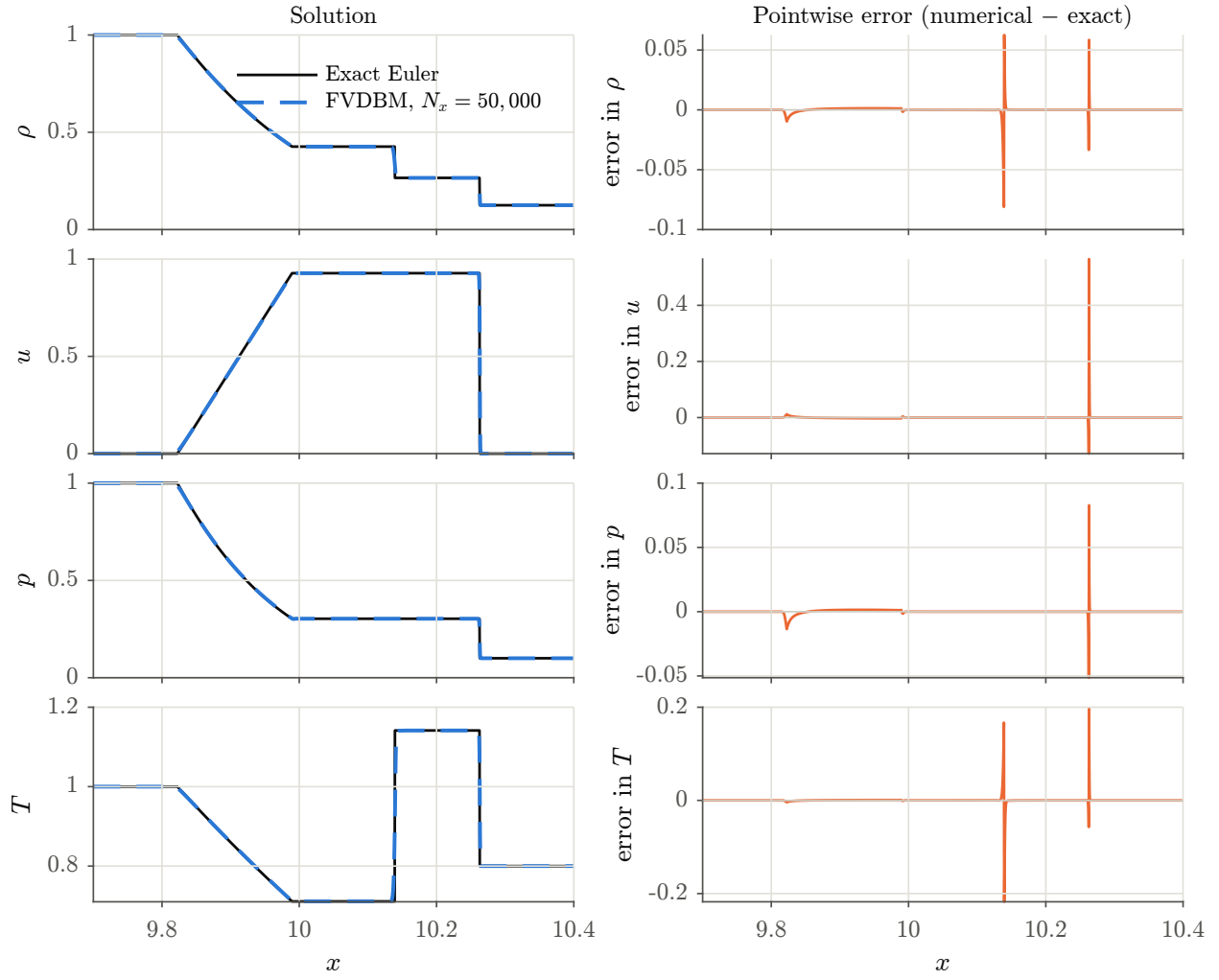
**Table 8:** Global  $L_2$  (RMS) errors and temperature peak excess.

| $N_x$  | $\rho$                | $u$                   | $p$                   | $T$                   | $T$ peak excess        |
|--------|-----------------------|-----------------------|-----------------------|-----------------------|------------------------|
| 3,125  | $3.16 \times 10^{-3}$ | $7.79 \times 10^{-3}$ | $3.08 \times 10^{-3}$ | $5.92 \times 10^{-3}$ | $3.88 \times 10^{-3}$  |
| 6,250  | $1.81 \times 10^{-3}$ | $8.06 \times 10^{-3}$ | $1.64 \times 10^{-3}$ | $4.65 \times 10^{-3}$ | $1.55 \times 10^{-3}$  |
| 12,500 | $1.39 \times 10^{-3}$ | $6.13 \times 10^{-3}$ | $1.14 \times 10^{-3}$ | $3.54 \times 10^{-3}$ | $-4.00 \times 10^{-5}$ |
| 25,000 | $1.00 \times 10^{-3}$ | $2.71 \times 10^{-3}$ | $7.20 \times 10^{-4}$ | $2.44 \times 10^{-3}$ | $-3.50 \times 10^{-6}$ |
| 50,000 | $7.77 \times 10^{-4}$ | $2.75 \times 10^{-3}$ | $5.15 \times 10^{-4}$ | $2.02 \times 10^{-3}$ | $1.04 \times 10^{-6}$  |

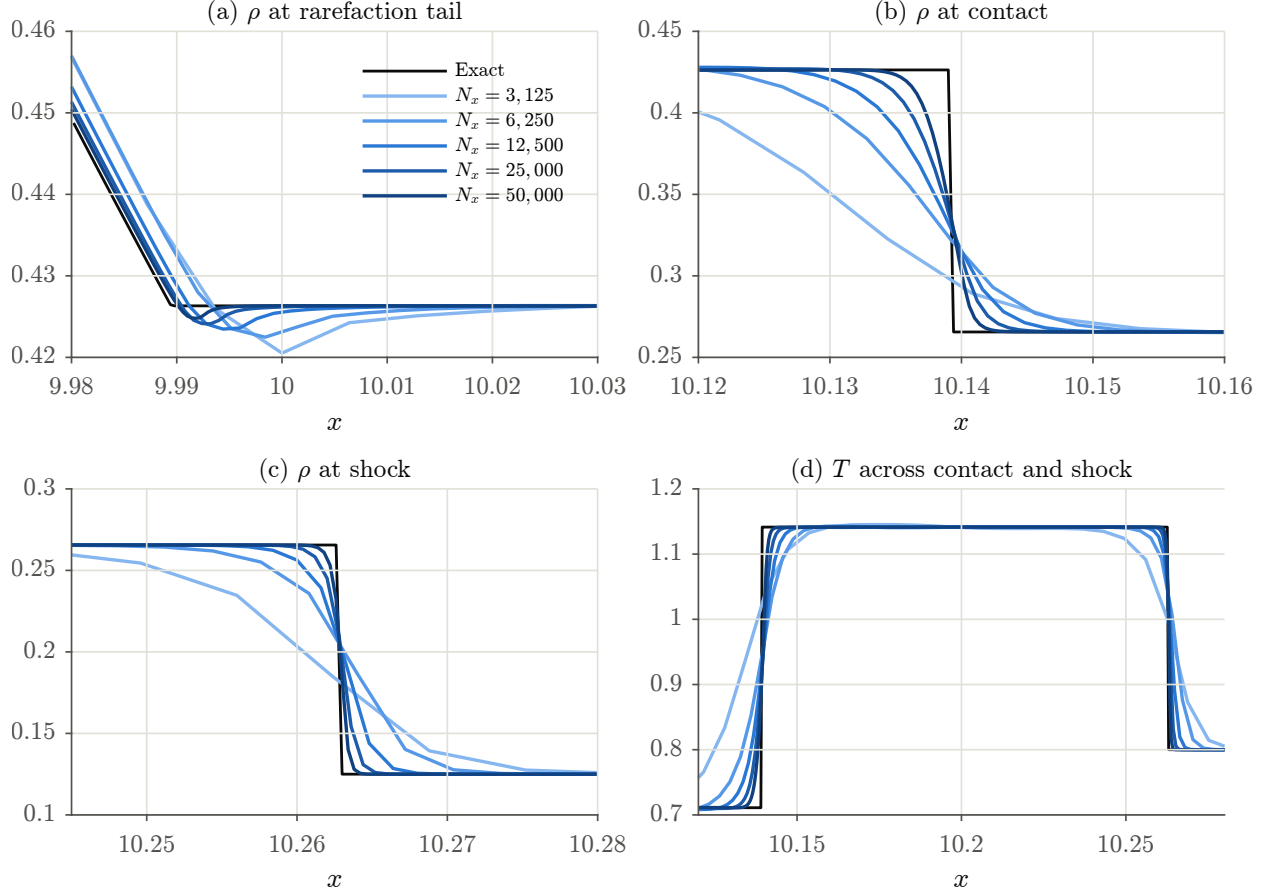
**Table 9:** Wave-region  $L_2$  errors,  $x \in [x_0 - 0.3, x_0 + 0.4]$ .

| $N_x$  | $\rho$                | $u$                   | $p$                   | $T$                   |
|--------|-----------------------|-----------------------|-----------------------|-----------------------|
| 3,125  | $1.69 \times 10^{-2}$ | $4.17 \times 10^{-2}$ | $1.65 \times 10^{-2}$ | $3.17 \times 10^{-2}$ |
| 6,250  | $9.65 \times 10^{-3}$ | $4.30 \times 10^{-2}$ | $8.74 \times 10^{-3}$ | $2.48 \times 10^{-2}$ |
| 12,500 | $7.43 \times 10^{-3}$ | $3.27 \times 10^{-2}$ | $6.11 \times 10^{-3}$ | $1.89 \times 10^{-2}$ |
| 25,000 | $5.36 \times 10^{-3}$ | $1.45 \times 10^{-2}$ | $3.85 \times 10^{-3}$ | $1.30 \times 10^{-2}$ |
| 50,000 | $4.15 \times 10^{-3}$ | $1.47 \times 10^{-2}$ | $2.75 \times 10^{-3}$ | $1.08 \times 10^{-2}$ |

**Interpretation.** All four variables converge. The least-squares  $L_1$  orders are 0.81–0.89, between the contact value  $\frac{2}{3}$  and the shock value 1 predicted in section 8.2. For the variables that jump



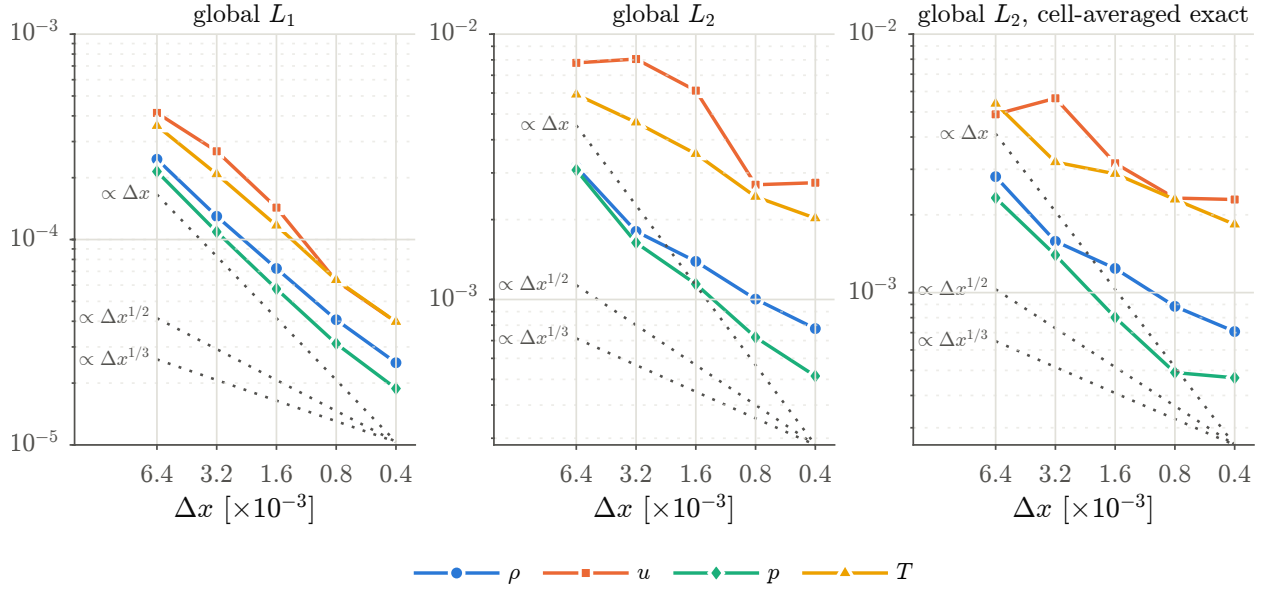
**Figure 8:** Finest grid ( $N_x = 50,000$ ) versus the exact Euler solution at  $t = 0.15$  (left) and the pointwise error (right), inside the error window.



**Figure 9:** Grid refinement at the three waves. Darker curves are finer grids. The shock (c) stays about three cells wide at every resolution. The contact (b) is wider in physical units and spans more cells as the grid is refined (table 11). The coarse-grid temperature overshoot in (d) disappears by  $N_x = 12,500$ .

**Table 10:** Observed orders. Columns 2–5: pairwise  $L_2$  orders (47) between successive grids. Then the least-squares  $L_2$  order over all five grids, the least-squares  $L_1$  order, and the least-squares wave-region  $L_2$  order.

|        | pairwise $L_2$ order |            |          |        | least-squares fit |       |            |
|--------|----------------------|------------|----------|--------|-------------------|-------|------------|
|        | 3125→6250            | 6250→12.5k | 12.5→25k | 25→50k | $L_2$             | $L_1$ | wave $L_2$ |
| $\rho$ | 0.809                | 0.376      | 0.473    | 0.369  | <b>0.490</b>      | 0.826 | 0.490      |
| $u$    | -0.048               | 0.395      | 1.179    | -0.025 | <b>0.458</b>      | 0.886 | 0.458      |
| $p$    | 0.911                | 0.517      | 0.667    | 0.485  | <b>0.634</b>      | 0.883 | 0.635      |
| $T$    | 0.349                | 0.394      | 0.537    | 0.269  | <b>0.403</b>      | 0.805 | 0.403      |



**Figure 10:** Convergence of the global  $L_1$  and  $L_2$  errors, with the  $L_2$  error also measured against the cell-averaged exact solution (right). Dotted lines are reference slopes.

across the contact ( $\rho$ ,  $T$ ) and for  $u$ , the  $L_2$  orders are 0.40–0.49, between the contact value  $\frac{1}{3}$  and the shock value  $\frac{1}{2}$ . Pressure, which does not jump at the contact, converges at 0.63, somewhat above  $\frac{1}{2}$ , because its faster-converging rarefaction error still makes up a visible share of the total. The ratio of the  $L_2$  order to the  $L_1$  order lies between 0.50 and 0.72, close to the value of one half predicted by (48). Temperature, which carries the largest relative contact jump, converges slowest. No pairwise order approaches two; the solution is not smooth, and formal second-order accuracy cannot be verified on this test.

#### 9.4 Error by wave: which feature limits convergence

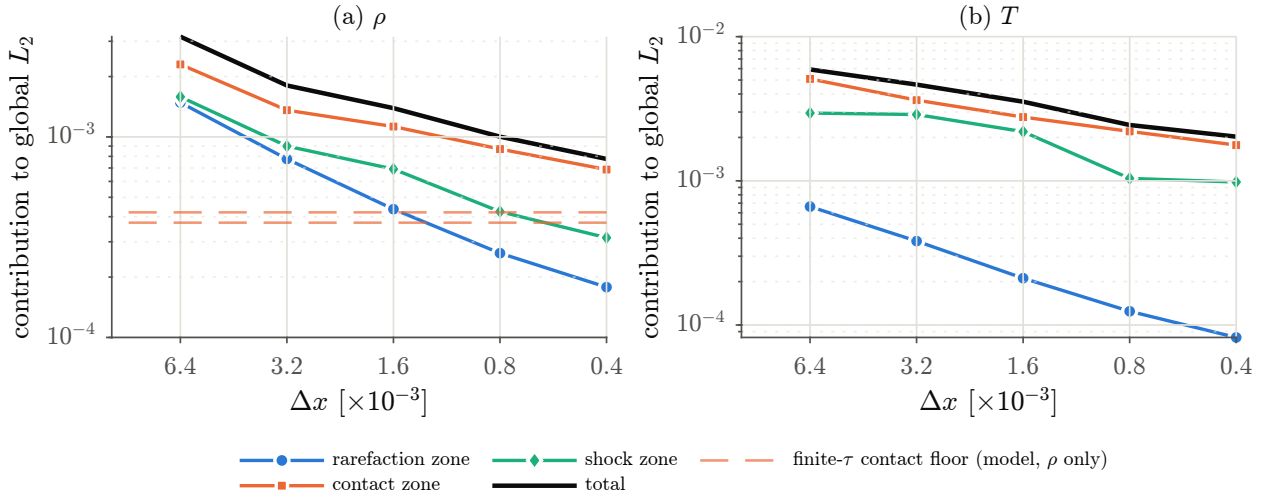
The global squared error is the sum of the contributions of the rarefaction, the contact and the shock. Figure 11 separates them using the zones  $[x_0 - 0.25, x_0 + 0.07]$ ,  $[x_0 + 0.07, x_0 + 0.20]$  and  $[x_0 + 0.20, x_0 + 0.40]$ . For density, the least-squares orders of the three contributions are 0.77 (rarefaction), 0.41 (contact) and 0.58 (shock). For temperature they are 0.76, 0.38 and 0.47. The contact is the largest contributor on every grid and has the slowest convergence, close to the theoretical  $\frac{1}{3}$ . It therefore controls the global  $\rho$  and  $T$  errors, and its share grows under refinement.

Table 11 and fig. 12 measure this directly. The shock thickness stays at 2.9–3.1 cells (physical

thickness order  $0.98 \approx 1$ ). This is the self-steepening behaviour of a captured shock. The contact thickness scales as  $\Delta x^{0.73}$ , close to the  $\Delta x^{2/3}$  law for a second-order scheme, so it spans more cells on finer grids: 4.3 cells at  $N_x = 3125$  and 9.3 cells at  $N_x = 50,000$ . The contact and shock position errors decrease steadily. The shock runs about 0.2 cells ahead of the exact position, and the contact lags by an amount that decreases faster than  $\Delta x$ .

**Table 11:** Measured contact and shock thickness (maximum-gradient definition, from  $\rho$ ) in physical units and in cells, and the position errors (numerical minus exact) of the mid-level crossings.

| $N_x$  | $\delta_c$           | $\delta_c/\Delta x$ | $\delta_s$           | $\delta_s/\Delta x$ | contact position error | shock position error  |
|--------|----------------------|---------------------|----------------------|---------------------|------------------------|-----------------------|
| 3,125  | $2.8 \times 10^{-2}$ | 4.33                | $1.9 \times 10^{-2}$ | 2.95                | $-8.4 \times 10^{-3}$  | $-1.8 \times 10^{-3}$ |
| 6,250  | $1.6 \times 10^{-2}$ | 5.13                | $9.4 \times 10^{-3}$ | 2.94                | $-2.2 \times 10^{-3}$  | $6.2 \times 10^{-4}$  |
| 12,500 | $9.7 \times 10^{-3}$ | 6.07                | $4.7 \times 10^{-3}$ | 2.95                | $-9.2 \times 10^{-4}$  | $3.4 \times 10^{-4}$  |
| 25,000 | $5.9 \times 10^{-3}$ | 7.42                | $2.5 \times 10^{-3}$ | 3.09                | $-3.7 \times 10^{-4}$  | $1.8 \times 10^{-4}$  |
| 50,000 | $3.7 \times 10^{-3}$ | 9.26                | $1.2 \times 10^{-3}$ | 3.05                | $-1.3 \times 10^{-4}$  | $7.9 \times 10^{-5}$  |

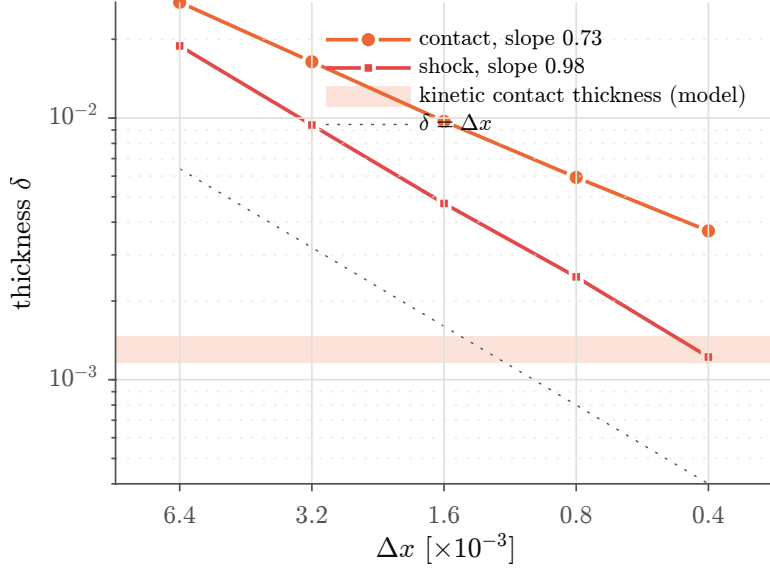


**Figure 11:** Contribution of each wave to the global  $L_2$  error for (a) density and (b) temperature. The dashed band in (a) is the finite- $\tau$  contact floor predicted by (26).

## 9.5 The velocity error and the sub-cell position of the shock

Velocity is the one variable whose  $L_2$  error does not decrease monotonically:  $L_2(u)$  rises slightly from  $N_x = 3125$  to 6250 and is flat from 25,000 to 50,000 (table 8). This matches the “velocity stops improving” observation recorded in earlier studies. The zone decomposition shows why. Between 93% (coarsest grid) and 99% (finest grid) of  $L_2(u)^2$  comes from the  $\pm 4$  cells around the shock, where  $u$  jumps by  $u^* = 0.93$ . The velocity is continuous across the contact, and the rarefaction contribution is four to ten times smaller than the shock contribution.

The contribution from the handful of cells that straddle the shock depends on *where the exact shock falls inside a cell*. The metric compares cell values with the exact solution sampled at cell centres. A shock near a face therefore produces one cell with an almost full-jump error, while a shock near a cell centre splits the error between two cells. With the exact shock at  $x_s = x_0 + 0.262823$ , its fractional position within its cell is 0.57, 0.13, 0.26, 0.53 and 0.06 on the five grids (table 12). Grouping the grids by this phase removes the irregularity. Between the two mid-cell grids ( $N_x =$



**Figure 12:** Measured contact and shock thickness versus grid spacing. The shaded band is the kinetic contact thickness (25) predicted for  $\tau = 5 \times 10^{-6}$ .

3125 and 25,000) the shock-zone  $L_2(u)$  converges with order 0.498. Between the two near-face grids (6250 and 50,000) the order is 0.513. Both match the theoretical  $\frac{1}{2}$  for a captured shock (fig. 13a). The flat 25,000  $\rightarrow$  50,000 velocity result is therefore a sampling artefact of comparing a point-sampled discontinuous reference with an  $\mathcal{O}(1)$ -per-cell error. It does not indicate a relaxation-time or time-step limit.

Averaging the exact solution over each cell, which is the natural reference for a finite-volume method, reduces the coarse-grid velocity error. It does not remove the phase dependence (fig. 13b, table 13), because the discrete shock profile itself depends on the shock’s sub-cell position. Two practical consequences follow. First,  $L_1(u)$ , which weights the few shock cells less heavily, converges monotonically with order 0.89 and is the more robust velocity metric. Second, velocity  $L_2$  orders should be quoted from least-squares fits over several grids, preferably with refinement ratios that sample different shock phases, rather than from a single pair of grids.

**Table 12:** Sub-cell position (phase) of the exact shock, the shock-zone ( $\pm 4\Delta x$ ) velocity error, and that error scaled by  $\Delta x^{1/2}$ , against point-sampled and cell-averaged exact solutions. A phase-independent error would give a constant scaled value.

| $N_x$  | phase | point-sampled exact   |                   | cell-averaged exact   |                   |
|--------|-------|-----------------------|-------------------|-----------------------|-------------------|
|        |       | shock-zone $L_2(u)$   | $/\Delta x^{1/2}$ | shock-zone $L_2(u)$   | $/\Delta x^{1/2}$ |
| 3,125  | 0.57  | $7.54 \times 10^{-3}$ | 0.0942            | $4.48 \times 10^{-3}$ | 0.0560            |
| 6,250  | 0.13  | $7.96 \times 10^{-3}$ | 0.1408            | $5.51 \times 10^{-3}$ | 0.0975            |
| 12,500 | 0.26  | $6.09 \times 10^{-3}$ | 0.1522            | $3.09 \times 10^{-3}$ | 0.0773            |
| 25,000 | 0.53  | $2.68 \times 10^{-3}$ | 0.0947            | $2.29 \times 10^{-3}$ | 0.0809            |
| 50,000 | 0.06  | $2.74 \times 10^{-3}$ | 0.1371            | $2.28 \times 10^{-3}$ | 0.1139            |

**Table 13:** Global  $L_2$  errors against the cell-averaged exact solution, with pairwise and least-squares orders (the last column is the least-squares  $L_1$  order).

|        | $N_x$     | $\rho$                | $u$                   | $p$                   | $T$                   |           |
|--------|-----------|-----------------------|-----------------------|-----------------------|-----------------------|-----------|
|        | 3,125     | $2.81 \times 10^{-3}$ | $4.90 \times 10^{-3}$ | $2.33 \times 10^{-3}$ | $5.38 \times 10^{-3}$ |           |
|        | 6,250     | $1.58 \times 10^{-3}$ | $5.65 \times 10^{-3}$ | $1.40 \times 10^{-3}$ | $3.20 \times 10^{-3}$ |           |
|        | 12,500    | $1.24 \times 10^{-3}$ | $3.17 \times 10^{-3}$ | $8.01 \times 10^{-4}$ | $2.88 \times 10^{-3}$ |           |
|        | 25,000    | $8.84 \times 10^{-4}$ | $2.32 \times 10^{-3}$ | $4.90 \times 10^{-4}$ | $2.29 \times 10^{-3}$ |           |
|        | 50,000    | $7.07 \times 10^{-4}$ | $2.29 \times 10^{-3}$ | $4.68 \times 10^{-4}$ | $1.84 \times 10^{-3}$ |           |
|        | 3125→6250 | 6250→12.5k            | 12.5→25k              | 25→50k                | $L_2$ fit             | $L_1$ fit |
| $\rho$ | 0.831     | 0.349                 | 0.488                 | 0.323                 | <b>0.482</b>          | 0.806     |
| $u$    | -0.206    | 0.836                 | 0.447                 | 0.018                 | <b>0.347</b>          | 0.838     |
| $p$    | 0.735     | 0.802                 | 0.708                 | 0.067                 | <b>0.614</b>          | 0.858     |
| $T$    | 0.751     | 0.151                 | 0.330                 | 0.321                 | <b>0.358</b>          | 0.772     |

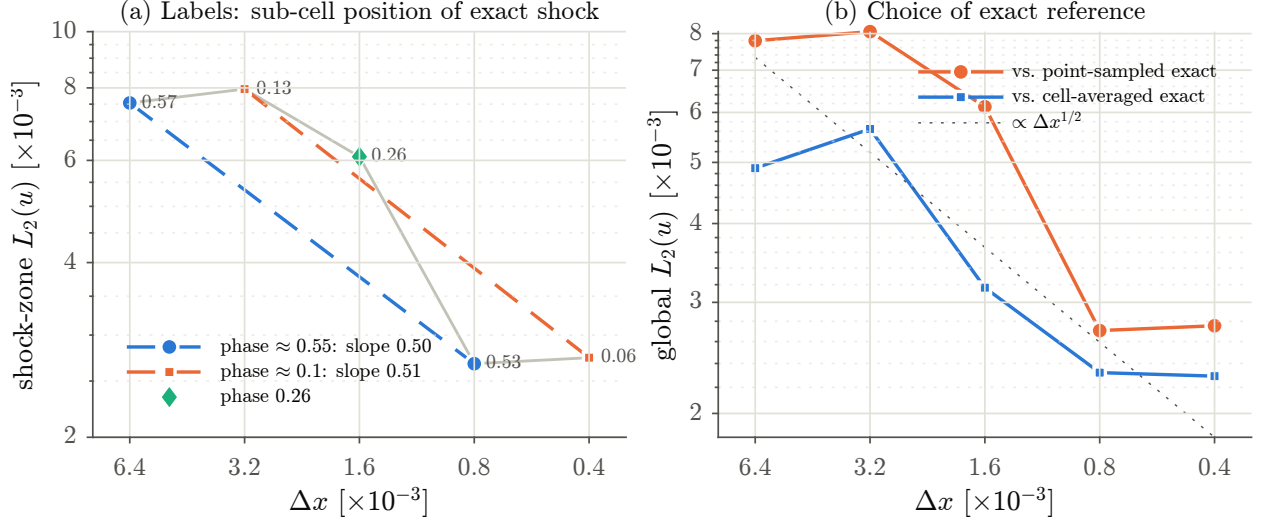
## 9.6 The role of the finite relaxation time

Section 4.5 predicts that at fixed  $\tau$  the contact cannot become thinner than  $\delta_c^{\text{kin}} = 1.2 \times 10^{-3} - 1.5 \times 10^{-3}$  for  $\tau = 5 \times 10^{-6}$ . The contact error therefore has a floor of  $3.7 \times 10^{-4} - 4.2 \times 10^{-4}$  in global  $L_2(\rho)$ . The measured contact thickness on the finest grid is  $3.7 \times 10^{-3}$ , about three times the kinetic value (fig. 12), and the contact-zone density error ( $6.8 \times 10^{-4}$ ) is still above the floor (fig. 11a). The present study is therefore discretization-dominated. Extrapolating the measured  $\Delta x^{0.73}$  contact thickness, the numerical and kinetic thicknesses become equal near  $\Delta x \approx 1 \times 10^{-4}$  ( $N_x \approx 2 \times 10^5$ ). Below that spacing, further refinement at  $\tau = 5 \times 10^{-6}$  would stop reducing the  $\rho$  and  $T$  errors.

For the earlier setting  $\tau = 5 \times 10^{-5}$  the kinetic contact is  $10^{1/2} \approx 3.2$  times thicker ( $3.7 - 4.6 \times 10^{-3}$ ). That is already equal to the numerical contact thickness at  $N_x = 50,000$ . This explains the plateau recorded in the project handoff for that setting (table 14):  $L_2(\rho)$  was  $1.130 \times 10^{-3}$  at  $N_x = 40,000$  and  $1.132 \times 10^{-3}$  at 50,000, whereas at  $\tau = 5 \times 10^{-6}$  the error keeps falling to  $7.77 \times 10^{-4}$ . The earlier runs also used a ten times larger time step ( $\Delta t = 1.25 \times 10^{-5}$ , CFL = 0.10), so the comparison does not separate  $\tau$  from  $\Delta t$ . It is, however, quantitatively consistent with the finite- $\tau$  contact floor.

**Table 14:** Global  $L_2(\rho)$  at the two relaxation times. The  $\tau = 5 \times 10^{-5}$  values are the September 13 results recorded in the project handoff (CFL = 0.10,  $\Delta t = 1.25 \times 10^{-5}$ ). The  $N_x = 50,000$  value at that  $\tau$  comes from the separate follow-on run in the same record.

| $N_x$  | $\tau = 5 \times 10^{-5}$ (handoff) | $\tau = 5 \times 10^{-6}$ (this study) | ratio |
|--------|-------------------------------------|----------------------------------------|-------|
| 3,125  | $3.244 \times 10^{-3}$              | $3.163 \times 10^{-3}$                 | 0.97  |
| 6,250  | $1.961 \times 10^{-3}$              | $1.806 \times 10^{-3}$                 | 0.92  |
| 12,500 | $1.594 \times 10^{-3}$              | $1.392 \times 10^{-3}$                 | 0.87  |
| 25,000 | $1.258 \times 10^{-3}$              | $1.003 \times 10^{-3}$                 | 0.80  |
| 40,000 | $1.130 \times 10^{-3}$              | —                                      | —     |
| 50,000 | $1.132 \times 10^{-3}$              | $7.766 \times 10^{-4}$                 | 0.69  |



**Figure 13:** (a) Shock-zone velocity error. Point labels give the sub-cell position of the exact shock. Grids with matching phase converge at order  $\approx \frac{1}{2}$ . (b) Global  $L_2(u)$  against point-sampled and cell-averaged exact solutions.

## 9.7 Limiter comparison

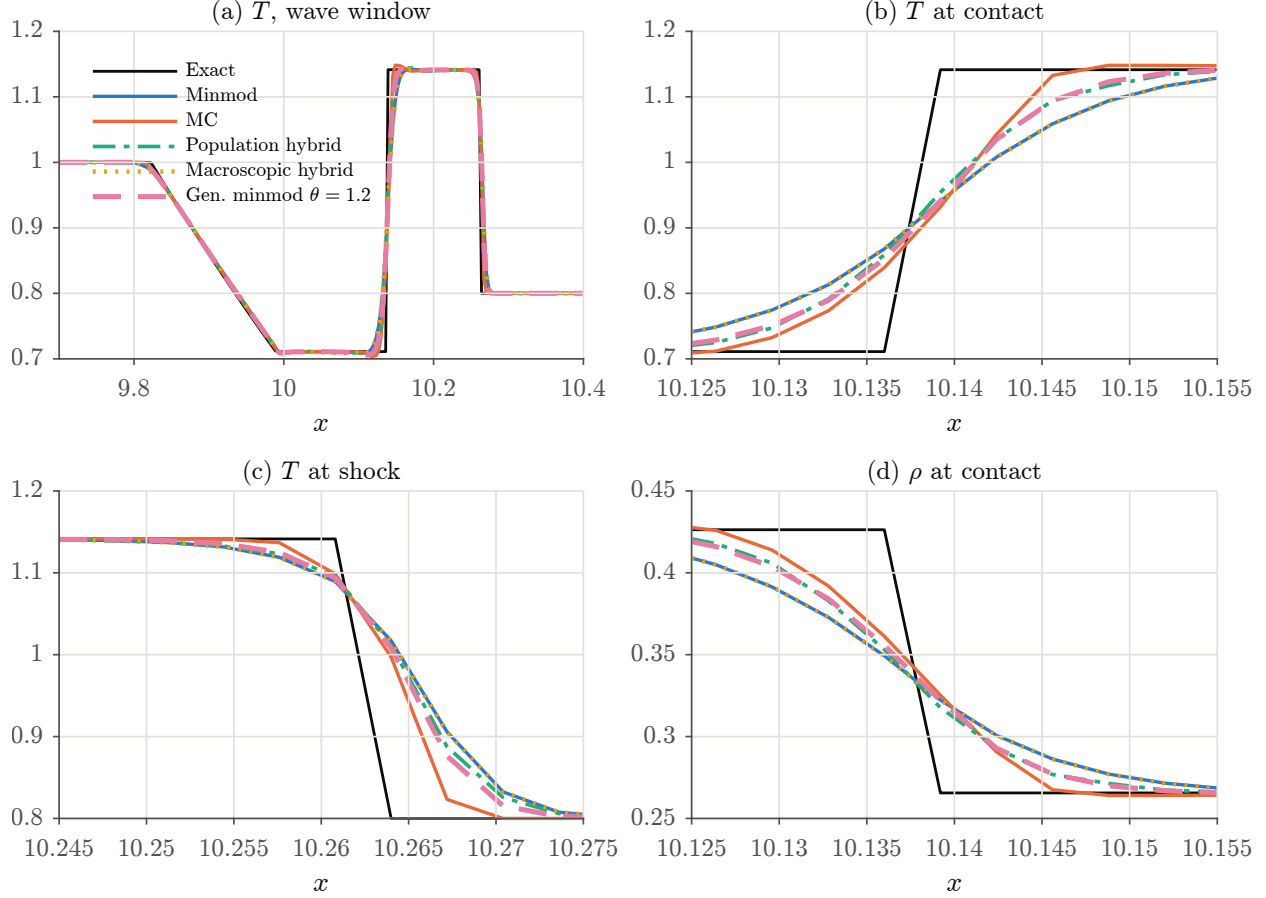
Table 15 and fig. 14 compare the five limiters developed during the project at  $N_x = 6250$ , under identical physics,  $\Delta t$ , grid and final time. The generalized-minmod case reproduces the convergence-study run at the same grid to all printed digits, which cross-checks the two independently written implementations.

**Table 15:** Limiter comparison,  $N_x = 6250$ ,  $\tau = 5 \times 10^{-6}$ , CFL = 0.025.

| Limiter                    | $L_1(\rho)$           | $L_1(u)$              | $L_1(p)$              | $L_1(T)$              | wave $L_1(T)$         | $T$ peak excess       |
|----------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Minmod                     | $1.68 \times 10^{-4}$ | $3.36 \times 10^{-4}$ | $1.42 \times 10^{-4}$ | $2.65 \times 10^{-4}$ | $7.57 \times 10^{-3}$ | $4.47 \times 10^{-4}$ |
| MC                         | $1.06 \times 10^{-4}$ | $2.07 \times 10^{-4}$ | $8.86 \times 10^{-5}$ | $1.68 \times 10^{-4}$ | $4.81 \times 10^{-3}$ | $6.72 \times 10^{-3}$ |
| Population hybrid          | $1.34 \times 10^{-4}$ | $2.89 \times 10^{-4}$ | $1.14 \times 10^{-4}$ | $2.16 \times 10^{-4}$ | $6.18 \times 10^{-3}$ | $2.59 \times 10^{-3}$ |
| Macroscopic hybrid         | $1.64 \times 10^{-4}$ | $3.27 \times 10^{-4}$ | $1.35 \times 10^{-4}$ | $2.66 \times 10^{-4}$ | $7.58 \times 10^{-3}$ | $6.16 \times 10^{-4}$ |
| Gen. minmod $\theta = 1.2$ | $1.30 \times 10^{-4}$ | $2.69 \times 10^{-4}$ | $1.09 \times 10^{-4}$ | $2.08 \times 10^{-4}$ | $5.94 \times 10^{-3}$ | $1.55 \times 10^{-3}$ |

The results reproduce, at the new  $\tau$  and CFL, the tradeoff identified earlier in the project:

- MC gives the smallest average errors but the largest temperature peak excess ( $6.7 \times 10^{-3}$ ). Its compressive slopes,  $\phi$  up to 2, overshoot at the contact, and the high-sensitivity populations of table 2 amplify the overshoot into  $T$ .
- Minmod has the smallest overshoot ( $4.5 \times 10^{-4}$ ) but the largest average errors.
- The macroscopic hybrid behaves almost exactly like minmod. Its shared troubled-cell mask evidently covers most of the cells in which minmod and MC would differ, so it rarely uses the MC slope where it matters.
- Generalized minmod with  $\theta = 1.2$  reduces all four global  $L_1$  errors by 20–23% relative to minmod, and reduces MC’s temperature peak excess by 77%.



**Figure 14:** Limiter comparison at  $N_x = 6250$ . Minmod and the macroscopic hybrid nearly coincide. MC is the sharpest and shows the largest temperature overshoot, just behind the contact (b).

## 9.8 The $\theta$ sweep

Table 16 and fig. 15 show the whole family from  $\theta = 1$  (identical to minmod) to  $\theta = 2$  (identical to MC). The average error decreases monotonically with  $\theta$ , and the temperature peak excess increases monotonically. The family therefore traces a single diffusion–overshoot tradeoff curve (fig. 15b). No  $\theta$  is best on both metrics. Measured along that curve,  $\theta = 1.2$  captures 59% of the  $L_1(T)$  reduction available between minmod and MC while incurring 18% of the overshoot increase. For comparison,  $\theta = 1.1$  gives 35%/7% and  $\theta = 1.3$  gives 74%/30%.  $\theta = 1.2$  is thus a reasonable compromise, not an optimum. The appropriate value depends on how peak temperature error is weighed against average error. On the convergence grids the peak excess at  $\theta = 1.2$  vanishes for  $N_x \geq 12,500$  (table 8), so the overshoot cost is a coarse-grid effect at this  $\tau$ . No fallback or retry occurred for any  $\theta$ .

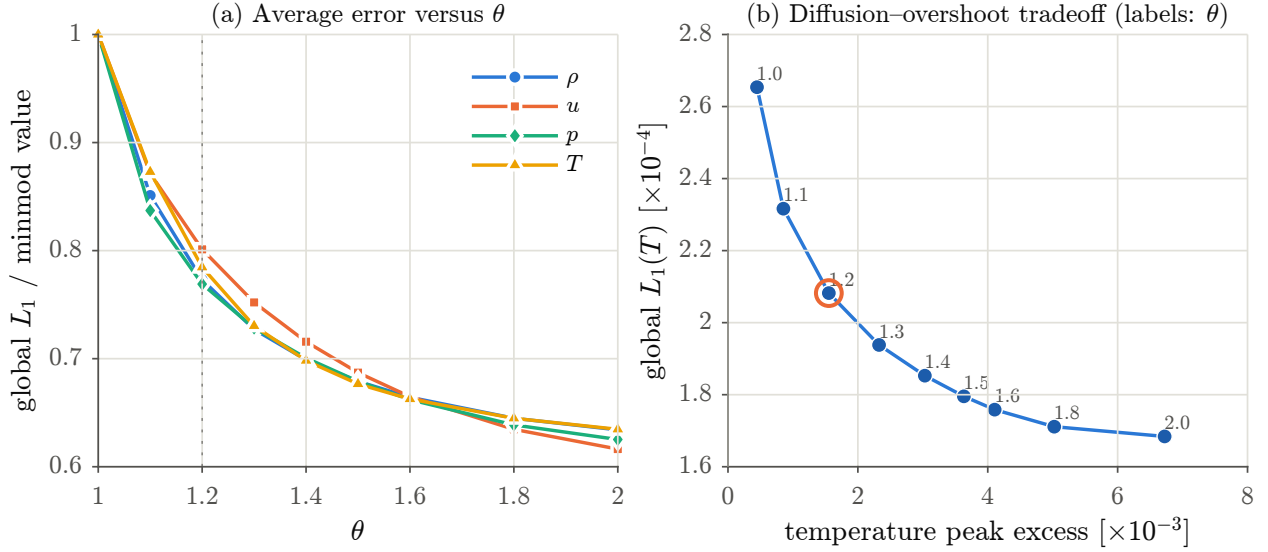
## 10 Summary

### 10.1 How the solver works, in one paragraph

The solver replaces the Euler equations with five linear advection equations, one per discrete velocity  $c_i \in \{0, \pm 1, \pm 3\}$ , coupled only through a local BGK relaxation toward the KT-D1V5 equilibrium.

**Table 16:**  $\theta$  sweep,  $N_x = 6250$ ,  $\tau = 5 \times 10^{-6}$ , CFL = 0.025. The last column counts limiter fallbacks.

| $\theta$    | $L_1(\rho)$           | $L_1(u)$              | $L_1(p)$              | $L_1(T)$              | wave $L_1(T)$         | $T$ peak excess       | fb |
|-------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----|
| 1.00        | $1.68 \times 10^{-4}$ | $3.36 \times 10^{-4}$ | $1.42 \times 10^{-4}$ | $2.65 \times 10^{-4}$ | $7.57 \times 10^{-3}$ | $4.47 \times 10^{-4}$ | 0  |
| 1.10        | $1.43 \times 10^{-4}$ | $2.93 \times 10^{-4}$ | $1.19 \times 10^{-4}$ | $2.32 \times 10^{-4}$ | $6.61 \times 10^{-3}$ | $8.53 \times 10^{-4}$ | 0  |
| <b>1.20</b> | $1.30 \times 10^{-4}$ | $2.69 \times 10^{-4}$ | $1.09 \times 10^{-4}$ | $2.08 \times 10^{-4}$ | $5.94 \times 10^{-3}$ | $1.55 \times 10^{-3}$ | 0  |
| 1.30        | $1.22 \times 10^{-4}$ | $2.53 \times 10^{-4}$ | $1.03 \times 10^{-4}$ | $1.94 \times 10^{-4}$ | $5.53 \times 10^{-3}$ | $2.33 \times 10^{-3}$ | 0  |
| 1.40        | $1.17 \times 10^{-4}$ | $2.40 \times 10^{-4}$ | $9.93 \times 10^{-5}$ | $1.85 \times 10^{-4}$ | $5.29 \times 10^{-3}$ | $3.03 \times 10^{-3}$ | 0  |
| 1.50        | $1.14 \times 10^{-4}$ | $2.31 \times 10^{-4}$ | $9.62 \times 10^{-5}$ | $1.80 \times 10^{-4}$ | $5.12 \times 10^{-3}$ | $3.63 \times 10^{-3}$ | 0  |
| 1.60        | $1.12 \times 10^{-4}$ | $2.23 \times 10^{-4}$ | $9.38 \times 10^{-5}$ | $1.76 \times 10^{-4}$ | $5.02 \times 10^{-3}$ | $4.11 \times 10^{-3}$ | 0  |
| 1.80        | $1.08 \times 10^{-4}$ | $2.13 \times 10^{-4}$ | $9.05 \times 10^{-5}$ | $1.71 \times 10^{-4}$ | $4.88 \times 10^{-3}$ | $5.02 \times 10^{-3}$ | 0  |
| 2.00        | $1.06 \times 10^{-4}$ | $2.07 \times 10^{-4}$ | $8.86 \times 10^{-5}$ | $1.68 \times 10^{-4}$ | $4.81 \times 10^{-3}$ | $6.72 \times 10^{-3}$ | 0  |



**Figure 15:** (a) Global  $L_1$  errors normalized by the minmod ( $\theta = 1$ ) values. (b) Temperature  $L_1$  error versus temperature peak excess. Each point is one  $\theta$ ; the solver's  $\theta = 1.2$  is circled.

That equilibrium is the unique five-population state whose moments reproduce the density, momentum and energy and the two Euler fluxes (proposition 1). Mass, momentum and energy are carried exactly by the collision operator and conservatively by the finite-volume transport. Each population is upwinded exactly, since upwinding is the Riemann solution for linear advection. Accuracy comes from a MUSCL reconstruction whose slope is limited by the generalized minmod, which is TVD for each population's transport when  $1 \leq \theta \leq 2$  (proposition 2). SSP-RK3 advances everything with a time step set by the relaxation time,  $\Delta t = \tau/4$ . This step resolves the BGK decay (stiffness exactly  $1/\tau$ ) and keeps the transport Courant number below 0.01 on every grid.

## 10.2 Principal findings

1. **Model.** The KT-D1V5 model reproduces the Euler equations exactly at leading order. Its  $\mathcal{O}(\tau)$  terms are not Navier-Stokes: the viscosity is about 3.5 times the BGK value, the dissipation is not Galilean invariant, and there is a spurious density-gradient energy flux (table 3).  $\tau$  should be regarded as a regularization that vanishes in the Euler limit.
2. **Contact physics of the model.** A contact moving at speed  $U$  diffuses with  $D_c = \tau(v_1^2 -$

$U^2)(v_2^2 - U^2)/[(b + 2)T]$ . At the Sod contact this is eight times weaker than at rest, and it would become anti-diffusive for  $|U| > v_1$ . At the level of this first-order analysis, the result limits the flow speeds for which the present velocity set is usable.

3. **Temperature sensitivity.**  $\partial T/\partial f_i = [(c_i - u)^2 + \eta_i^2 - bT]/(b\rho)$  reaches about 7 for the fast counter-moving population in the post-shock state. This explains why limiter-induced overshoot appears first, and most strongly, in  $T$ .
4. **Convergence.** With  $\tau = 5 \times 10^{-6}$  and CFL = 0.025 on five grids up to  $N_x = 50,000$ , all runs were clean (no retries, fallbacks or repairs) and conservative to round-off. The fitted orders are  $L_1 \approx 0.81\text{--}0.89$  and  $L_2 \approx 0.40\text{--}0.63$ , which is the correct behaviour for a second-order method on a solution with a shock and a contact. The contact is the slowest-converging feature (thickness  $\propto \Delta x^{0.73}$ , close to the theoretical  $\Delta x^{2/3}$ ). The shock stays three cells wide.
5. **Velocity anomaly explained.** The non-monotone  $L_2(u)$  comes from the sub-cell position of the exact shock relative to the grid. Grids with matching phase converge at order 0.50–0.51.  $L_1(u)$  converges monotonically.
6. **Relaxation time.** At  $\tau = 5 \times 10^{-6}$  the grids used here remain discretization-dominated. The kinetic contact floor is predicted to matter only near  $N_x \approx 2 \times 10^5$ . At  $\tau = 5 \times 10^{-5}$  the floor is reached near  $N_x \approx 5 \times 10^4$ , consistent with the plateau seen in the earlier study.
7. **Limiter.**  $\theta = 1.2$  lowers every global  $L_1$  error by 20–23% relative to minmod and cuts MC’s temperature overshoot by 77%. The  $\theta$  sweep shows a single monotone tradeoff:  $\theta = 1.2$  buys 59% of the attainable  $L_1(T)$  improvement for 18% of the overshoot cost. It is a defensible compromise, not a unique optimum.

### 10.3 Limitations and recommended next steps

- Every convergence run used the same  $\Delta t$ , so temporal error was not measured. A short  $\Delta t$ -halving test at  $N_x = 50,000$  would confirm the expectation that it is negligible.
- Formal second-order accuracy should be verified on a smooth problem (for example an isentropic or small-amplitude acoustic pulse), because Sod cannot show it.
- The finite- $\tau$  floor analysis is a first-order Chapman–Enskog estimate. A  $\tau$  study at fixed  $\Delta t$  and a fine grid would test it directly.
- Velocity convergence should be reported from  $L_1$  norms or from phase-varied grid sequences, to avoid the shock-phase artefact.
- The contact-diffusion result implies that a two-dimensional extension needs a velocity set whose slowest speed exceeds the expected flow speeds, or a higher-order (Navier–Stokes-level) equilibrium [5, 10].

## A Reproducibility

All numbers and figures in section 9 are generated by scripts in the research folder:

- `run_d1v5_gminmod_L2_convergence.m`: five-grid study. Each grid is cached in `results/runs/` with  $\theta$ ,  $\tau$ , CFL and  $N_x$  in the file name. `run_d1v5_gminmod_L2_convergence(Nx)` runs a single grid, which allows several grids to run in parallel.

- `d1v5_sodshock_limiter_history_comparison.m`: five-limiter comparison at  $N_x = 6250$ ,  $\tau = 5 \times 10^{-6}$ ,  $\text{CFL} = 0.025$ .
- `report_d1v5_math/scripts/run_d1v5_theta_sweep.m`:  $\theta$  sweep using the unmodified solver.
- `report_d1v5_math/scripts/verify_kt_d1v5_theory.m`: numerical checks of every analytical result in sections 3 and 4.
- `report_d1v5_math/scripts/make_theory_figures.m` and `make_results_figures.m`: all figures and the tables in section 9.

The earlier  $\tau = 5 \times 10^{-5}$  and  $\tau = 5 \times 10^{-6}/\text{CFL} = 0.05$  result files are preserved unchanged in `results/backup_2026-09-16_tau5e-6_study/`.

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